Joint Beamforming and Data Stream Allocation for Non-Coherent Joint Transmission

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***Abstract*—This paper addresses the joint beamforming and data stream allocation (DSA) optimization problem for non-coherent joint transmission (NCJT). A critical yet neglected issue in NCJT beamforming is the tightly related DSA, which involves determining the number of streams transmitted from access points (APs) to their serving user equipments (UEs) according to the channel quality, so that the weighted sum-rate (WSR) can be maximized. However, since the integer number of streams directly determines the dimensions of beamformers, the joint optimization problem is mixed-integer and nonconvex with tightly coupled decision variables, making it NP-hard. To solve this problem, we first fix the DSA variables and propose a distributed and reduced weighted minimum mean square error (WMMSE) beamforming algorithm, called distributed RWMMSE, by leveraging the low-dimensional subspace property of the beamformer obtained by the traditional WMMSE. The distributed RWMMSE achieves the same WSR as the traditional WMMSE but with significantly lower computational complexity (scaling linearly with the number of AP antennas) and reduced interaction cost. Building on this, a joint beamforming and DSA optimization algorithm, named RWMMSE-LSA, is proposed by decoupling the decision variables through introduced stream indicator matrices. The RWMMSE-LSA optimizes beamformers and DSA via the distributed RWMMSE and linear programming, respectively, both of which have closed-form solutions. Simulations validate substantial performance gain of our proposed algorithms over existing alternatives in computational and interaction costs.**

***Index Terms*—Non-coherent joint transmission, beamforming, data stream allocation, mixed-integer programming.**

Received 28 April 2025; revised 19 August 2025; accepted 2 September 2025. Date of publication 10 September 2025; date of current version 23 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62231019, and in part by the Science and Technology Commission Foundation of Shanghai under Grant 24DP1500704. An earlier version of this paper was presented in part at the 2023 IEEE Global Communications Conference (GLOBECOM), Kuala Lumpur, Malaysia, December, 2023 [DOI: 10.1109/GLOBECOM54140.2023.10437404]. The associate editor coordinating the review of this article and approving it for publication was S. Ahmed.
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Digital Object Identifier 10.1109/TCOMM.2025.3608318

I. Introduction

THE proliferation of advanced mobile applications has led to a dramatic increase in global mobile data traffic, which is expected to grow by a factor of 2.5 by 2030 [2]. This surge in data demand has significantly increased the need for gigabit-per-second data rates and extremely high spectral efficiency to maintain quality of service. To meet these requirements, one effective approach is joint transmission (JT), where user equipments (UEs) are jointly served by multiple access points (APs) connected via fronthaul links. With evidence highlighting its potential to mitigate interference and thereby to improve spectral efficiency, JT has been envisioned as a promising technique to empower 5G and upcoming 6G systems [3].

Depending on whether the cooperative APs transmit the same signal to their target UE or not, JT can be classified into coherent JT (CJT) and non-coherent JT (NCJT), as illustrated in Fig. 1. In CJT, multiple APs collaborate as a virtual multiple-input-multiple-output (MIMO) system and transmit identical signal to their target UE, thereby exploiting higher coherent gain. However, CJT requires strict synchronization and substantial fronthaul capacity among the APs [4], [5], posing significant challenges for practical implementation, particularly when APs are equipped with a large number of antennas [6], [7]. In contrast, NCJT allows the cooperative APs to send distinct signals to the same target UE, offering an efficient solution to the synchronization and fronthaul load issues encountered in CJT [8]. NCJT has been shown to outperform CJT under conditions of limited fronthaul capacity [4], [8]. To date, NCJT has been widely used in 3GPP radio access network (RAN), including cloud-RAN (C-RAN) [4], coordinated multipoint (CoMP) [9], dense small cell networks [10], and cell-free networks [8], [11].

Although widely implemented, NCJT presents unique challenges for beamforming design that remain largely unexplored in the literature, hindering its full potential. On the one hand, beamforming design is intrinsically related to the number of data streams transmitted from APs to UEs, as this number directly determines the dimensions of the beamforming matrices. On the other hand, intuitively, APs with better channel conditions should transmit more data streams to their target UE. Therefore, beamforming should be jointly optimized with data stream allocation (DSA) in NCJT to maximize communication efficiency. In contrast, CJT does not face this issue, since the serving APs transmit the same signal to the target UE, implying that the optimal number of data streams matches

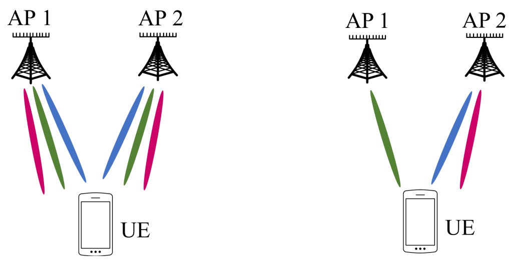


Fig. 1. An exemplary illustration of CJT and NCJT, where different colored beams represent different data streams.

the number of UE receive antennas [11] (UEs typically have much fewer antennas than APs). As a result, unlike CJT that mainly involves pure beamforming design, NCJT requires the joint optimization of beamforming and DSA, making it a more complex but potentially more rewarding design challenge.

While several studies have investigated NCJT, their proposed approaches exhibit limitations in both beamforming design and DSA. First, current NCJT beamforming methods encounter significant computational and interaction challenges [4], [11], [12], [13]. Their computational complexity scales at least cubically with the number of antennas per AP, denoted by M, and the interaction overhead grows linearly with M, due to the exchange of raw channel state information (CSI) and beamforming matrices over fronthaul links. These demands impose substantial burdens on both computational resources and fronthaul capacity in massive MIMO systems, where M can reach hundreds or even thousands [14]. Second, prior works typically restrict each AP to transmitting a single data stream to each single-antenna UE, and focus instead on AP-UE pairing, i.e., addressing problems such as user scheduling [11], user pairing [12], and user association [13], rather than comprehensive DSA optimization. Notably, AP-UE pairing is a reduced case of DSA, as selecting an AP for a UE is equivalent to assigning a non-zero number of data streams.

To address these limitations, this paper explores the joint optimization of beamforming and DSA in NCJT, aiming at enhancing the weighted sum-rate (WSR) of multi-user MIMO (MU-MIMO) networks with reduced computational complexity and interaction overhead.

A. Related Works

1) Beamforming Design: Existing NCJT beamforming algorithms demand massive raw CSI interaction and suffer from high computational cost. In [4], the successive convex approximation (SCA) method was employed to relax the nonconvex sum-rate maximization problem in NCJT into a second-order cone programming (SOCP) problem, which was solved by the CVX toolbox [15]. A similar SCA-based beamforming approach was adopted in [12] for non-orthogonal multiple access systems. Following the SCA framework, [10] used the inner approximation (InAp) method to relax the nonconvex WSR maximization problem to a convex approximation subproblem, which was then solved by the alternating direction method of multipliers (ADMM). However, the computational complexity of the approaches adopted in [4], [10], and [12] scales cubically with the total number of antennas across all APs.

Note that the above SCA-based algorithms in [4], [10], and [12] have no closed-form solution. In order to obtain a closed-form solution at each iteration, the authors in [11] utilized the fractional programming (FP) approach to optimize the beamformers for WSR maximization under the special case that each UE is equipped with a single antenna. The FP-based algorithm in [11] has much lower complexity than the SCA-based algorithms, but still suffers from a $\mathcal{O}\left(M^3\right)$ computational complexity that is onerous especially when massive antennas are deployed. Another efficient algorithm that has closed-form beamforming solutions at each iteration is the weighted minimum mean square error (WMMSE) algorithm [16], which has been widely employed in CJT [17], [18]. However, its applicability to NCJT remains an open problem [4], [10], due to the unique structure of the sum of signal covariance in its communication rate expression. In Section III-A, we will establish that through proper reformulation, the WMMSE framework can indeed be extended to NCJT systems.

Apart from high computational complexity, the aforementioned SCA-based [4], [12], FP-based [11], WMMSE-type algorithms [16], [17], [18], and even heuristic zero forcing (ZF) precoding [13] all necessitate the interaction of high-dimensional raw channel and beamforming matrices. Since both raw CSI and beamformers scale with the number of transmit antennas M, these schemes could result in a prohibitive burden on the fronthaul links, especially in massive MIMO systems.

Although significant efforts have been made to reduce the interaction cost for beamforming design, these methods still suffer from performance loss due to partial CSI acquisition, and incur high computational complexity. Specifically, the works [19], [20] developed local minimum mean square error (L-MMSE) beamforming by omitting terms involving CSI from other APs in the traditional MMSE approach to avoid inter-AP CSI exchange, which inherently results in sum-rate loss. Iterative bi-directional training (IBT) methods were proposed in [21] and [22] for distributed beamforming, which allow each AP to obtain necessary CSI from dedicated uplink pilots. However, the IBT algorithms introduce significant processing delay due to the iterative nature, and necessitate long uplink pilot sequences to maintain performance, with computational complexity being $\mathcal{O}(M^3)$. In addition, [23], [24], [25] opted to exchange interference terms instead of raw channel matrices. However, the interference graph in practice may not align with the fronthaul graph, posing new challenges in exchanging interference terms among APs. Furthermore, the algorithms in [23], [24], and [25] exhibit a complexity of $\mathcal{O}(M^{3.5})$.

2) Data Stream Allocation: Note that existing studies in NCJT focus on networks where each UE is equipped with only one antenna [4], [10], [11], [12], [13], so each serving AP transmits only one data stream to the UE. However, UEs are generally equipped with multiple antennas in practice, thus can receive multiple data streams simultaneously. Given the large number of transmit antennas [14], there is significant potential to improve the WSR by properly allocating data streams from the APs to their serving UEs.

Indeed, DSA for WSR maximization constitutes a mixed-integer programming problem and is classified as NP-hard. Due to its difficulty, only a limited number of studies have explored DSA, and these are restricted to the single-AP scenario. In [26], the channels corresponding to the equivalent channel vectors with either the highest M eigenvalues or the highest M channel norms were selected for transmission. In [27], [28], and [29], greedy-based DSA approaches were developed. In each step, an additional data stream is allocated to the UE based on the criterion of increasing the sum-rate most [27], [28] or decreasing the transmission power most [29]. Integrated with the DSA approaches, block diagonalization [27], [29] and ZF [28] methods were exploited to design the beamforming vectors for the single AP. Although these greedy-based DSA algorithms perform remarkably close to optimum [30], they involve huge computational cost, especially when the number of data streams surges.

B. Contributions

To address the high computational complexity and interaction overhead in existing beamforming designs, as well as the limitation of current DSA methods, this paper investigates joint beamforming and DSA optimization for NCJT, with the aim of maximizing the WSR of MU-MIMO networks while significantly reducing both computational and interaction costs. To the best of our knowledge, this is the first work that considers joint beamforming and DSA for NCJT. Despite promising, it comes with the following three major challenges:

1) Beamforming design and DSA are *intrinsically coupled* and hard to be jointly optimized, since the dimension of the beamformer is determined by the number of data streams.
2) Even with a fixed number of data streams, the pure beamforming optimization problem remains NP-hard. Existing NCJT beamforming algorithms require high computational and interaction costs, and are designed for networks with single-antenna UEs [4], [10], [11], [12], [13].
3) Since reducing CSI exchange and computational cost generally degrades the system performance [13], [20], it is challenging to significantly reduce the interaction and computational costs without sacrificing the WSR.

Due to the above challenges, we first fix the number of streams, and study a pure beamforming approach that reduces both interaction and complexity without compromising the WSR. Building upon this, we further explore the joint beamforming and DSA problem for WSR maximization in NCJT. The main contributions are summarized as follows:

1) **Distributed and Reduced WMMSE Beamforming for NCJT:** We first demonstrate the applicability of the WMMSE algorithm [16], originally developed for CJT, to the pure beamforming problem in NCJT. Based on this, we propose a distributed and reduced WMMSE algorithm, named distributed RWMMSE, by exploiting the low-dimensional subspace structure of the beamforming matrices obtained through WMMSE. The proposed algorithm maintains the same WSR as the original WMMSE while significantly reducing both computational and interaction costs for large-scale antenna array systems. Specifically, it replaces the update of the original beamformer with its low-dimensional substitution, which requires only $\{\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H\}$ rather than the raw channel matrices $\{\bar{\mathbf{H}}_i\}$ as in the original WMMSE, where $\bar{\mathbf{H}}_i$ denotes the channel matrices between each AP i and all UEs. This substitution results in a computational complexity that scales linearly with M, yielding a substantial reduction from the $\mathcal{O}(M^3)$ complexity of the original WMMSE, and the interaction cost also scales down.
2) **Joint Beamforming and DSA for NCJT:** Building on the proposed distributed RWMMSE algorithm, we further address the joint beamforming and DSA optimization problem, which remains challenging due to its mixed-integer nature and the high coupling of the decision variables. To tackle this issue, we introduce auxiliary stream indicator matrices as substitutes for the original DSA variables, which successfully decouple the decision variables. By leveraging the unique quadratic-linear property of these indicator matrices, the quadratic and integer optimization problem is equivalently transformed into a linear and continuous optimization problem, where the beamforming and indicator matrices are optimized via distributed RWMMSE and linear programming, respectively, both with closed-form solutions. The proposed joint beamforming and DSA algorithm, termed RWMMSE-LSA, maintains computational complexity comparable to that of distributed RWMMSE, and is guaranteed to converge to the stationary points of the equivalent continuous optimization problem.
3) Extensive simulations are conducted to verify the effectiveness of the proposed algorithms. The distributed RWMMSE algorithm achieves the same WSR as the traditional WMMSE, but requires much less convergence time, which aligns with our theoretical analysis. Furthermore, the proposed joint beamforming and DSA algorithm outperforms the pure beamforming algorithm with evenly assigned DSA in terms of WSR, highlighting the benefits of incorporating DSA optimization.

The remainder of the paper is organized as follows. Section II describes the system model and problem formulation for WSR maximization. In Section III, we present the distributed RWMMSE algorithm, along with the interaction cost and computational complexity. Subsequently, we introduce the RWMMSE-LSA algorithm for joint beamforming and DSA in Section IV. Simulation results are provided in Section V, followed by concluding remarks in Section VI.

Notations: We use bold lower and upper case letters to represent vectors and matrices, i.e., $\mathbf{a}$, $\mathbf{A}$, respectively. For a matrix $\mathbf{A}$, $\mathbf{A}^T$, $\mathbf{A}^H$, $\mathbf{A}^{-1}$, $\mathrm{Tr}(\mathbf{A})$, $\mathbf{A}^\dagger$, and $\|\mathbf{A}\|_{\mathrm{F}}$ signify its transpose, conjugate transpose, inverse, trace, pseudo inverse, and Frobenius norm, respectively. The notation $\mathbf{A} \succ \mathbf{0}$ indicates positive definite for matrix $\mathbf{A}$. The notation

TABLE I
SUMMARY OF KEY NOTATIONS

Symbol	Description	Symbol	Description
α_k	Weighting factor for UE k	$\tilde{\mathbf{H}}_i$, $\bar{\mathbf{H}}_i$	Channel matrix between APs $\mathcal{I}_k$ and all UEs
$D_{i,k}$, $D_{(k)}$	Number of data streams transmitted from AP i to UE k and from APs $\mathcal{I}_k$ to UE k	$\tilde{\mathbf{H}}_{\mathcal{I}_k}$	Channel matrices between AP i and its serving UEs, and between AP i and all UEs
D_{total}	Number of useful data streams of all UEs	$\mathbf{L}_{i,k}$, $\mathbf{L}_k$	Stream indicator matrices at AP i and at APs $\mathcal{I}_k$ for UE k
I, $\mathcal{I}$	Total number of APs and set of all APs	$\mathbf{N}_k$	Interference-plus-noise covariance matrix at UE k
K, $\mathcal{K}$	Total number of UEs and set of all UEs	$\mathbf{P}_{i,k}$, $\bar{\mathbf{P}}_{i,k}$	Beamformer and virtual beamformer at AP i for UE k
$\mathcal{K}_i$	Set of UEs served by AP i	$\mathbf{P}_k$, $\bar{\mathbf{P}}_k$	Beamformer and virtual beamformer at APs $\mathcal{I}_k$ for UE k
$\mathcal{I}_k$, $\lvert\mathcal{I}_k\rvert$	Set of APs serving UE k and its cardinality	$\tilde{\mathbf{P}}_i$	Beamformer at AP i for its serving UEs
M_i, $M_{(k)}$	Number of antennas at AP i, and at APs $\mathcal{I}_k$	$\mathbf{Q}_{i,k}$, $\bar{\mathbf{Q}}_i$	Left singular matrices of $\mathbf{H}_{i,k}$ and $\bar{\mathbf{H}}_i^H$
N_k, N_{total}	Number of antennas at UE k and at all UEs	$\mathbf{U}_k$, $\mathbf{W}_k$	Auxiliary matrices introduced by WMMSE
$P_{\max,i}$	Power budget of AP i	$\mathbf{V}_{i,k}$, $\bar{\mathbf{V}}_i^H$	Right singular matrices of $\mathbf{H}_{i,k}$ and $\bar{\mathbf{H}}_i^H$
$\mathcal{L}$	Set of stream indicators	$\tilde{\mathbf{V}}_i$	Effective channel matrix between AP i and its serving UEs
$\mathcal{P}$, $\bar{\mathcal{P}}$	Sets of beamformers and virtual beamformers	$\mathbf{X}_{i,k}$, $\check{\mathbf{X}}_{i,k}$	Low-dimensional substitutions for $\mathbf{P}_{i,k}$
$\mathbf{H}_{i,k}$	Channel matrix between AP i and UE k	$\bar{\mathbf{X}}_{i,k}$, $\bar{\mathbf{X}}_{\mathcal{I}_k}$	Low-dimensional substitutions for $\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}$ and for $\bar{\mathbf{P}}_k\mathbf{L}_k$
$\mathbf{H}_k$	Channel matrix between APs $\mathcal{I}_k$ and UE k	$\tilde{\mathbf{X}}_i$, $\tilde{\mathbf{X}}_{\mathcal{I}_k}$	Low-dimensional substitutions for $\tilde{\mathbf{P}}_i$ and for $\mathbf{P}_k$

$\text{blkdiag}(\mathbf{A}_1,\ldots,\mathbf{A}_n)$ denotes a block diagonal matrix with matrices $\mathbf{A}_1,\ldots,\mathbf{A}_n$. The real part of a complex number a is denoted as $\text{Re}(a)$. The notation $\mathbf{x}\sim\mathcal{CN}(\mathbf{m},\mathbf{R})$ denotes a circularly symmetric complex Gaussian (CSCG) vector with mean $\mathbf{m}$ and covariance $\mathbf{R}$. The binary space, complex space, real number space, and non-negative integer space are denoted by $\mathbb{B}$, $\mathbb{C}$, $\mathbb{R}$, and $\mathbb{N}$, respectively.

II. System Model and Problem Formulation

This section will clarify the system model and formulate the joint beamforming and DSA problem.

A. System Model

Consider a downlink user-centric massive MU-MIMO network that comprises I geographically distributed APs and K UEs, denoted by $\mathcal{I}=\{1,\ldots,I\}$ and $\mathcal{K}=\{1,\ldots,K\}$, respectively. Each AP i is equipped with M_i transmit antennas, serving a subset of UEs $\mathcal{K}_i\subseteq\mathcal{K}$. Each UE k is equipped with N_k receive antennas, and is cooperatively served by a subset $\mathcal{I}_k$ of APs with $\mathcal{I}_k\subseteq\mathcal{I}$. All the APs are connected through fronthaul links to a central processing unit (CPU), which collects CSI from APs and handles necessary signal processing for beamforming and DSA.

In this paper, we consider the scenario that APs cooperate in the NCJT mode, i.e., each AP $i\in\mathcal{I}_k$ transmits distinct signals $\mathbf{s}_{i,k}\in\mathbb{C}^{D_{i,k}\times 1}$ to its target UE k, where $D_{i,k}$ is the number of data streams and $\mathbb{E}[\mathbf{s}_{i,k}\mathbf{s}_{i,k}^H]=\mathbf{I}$. Note that while most papers assume fixed $D_{i,k}$, this paper treats $D_{i,k}$ as optimizable parameters. Specifically, we consider the allocation of the number of streams transmitted by serving APs $\mathcal{I}_k$ to UE k as variables that can be adjusted based on channel quality to improve the overall communication performance. Denote by $\mathbf{H}_{i,k}\in\mathbb{C}^{N_k\times M_i}$ the channel between AP i and UE k.[1] The received signal at UE k can be expressed as

$$\mathbf{y}_k=\sum_{i\in\mathcal{I}_k}\mathbf{H}_{i,k}\mathbf{P}_{i,k}\mathbf{s}_{i,k}+\sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\mathbf{P}_{j,l}\mathbf{s}_{j,l}+\mathbf{z}_k, \quad (1)$$

where $\mathcal{K}_{-k}\triangleq\mathcal{K}\backslash\{k\}$, $\mathbf{P}_{i,k}\in\mathbb{C}^{M_i\times D_{i,k}}$ is the beamforming matrix between AP i and UE k, $\mathbf{z}_k\sim\mathcal{CN}\left(\mathbf{0},{\sigma_k}^2\mathbf{I}\right)$ is the additive white Gaussian noise (AWGN) at UE k. Note that the first term in (1) is the useful signals transmitted from APs $\mathcal{I}_k$ to UE k, while the second term denotes the inter-user interference. Based on (1), the achievable data rate of UE k can be computed as [4], [10][2]

$$R_k=\log\det\left(\mathbf{I}+\left(\sum_{i\in\mathcal{I}_k}\mathbf{H}_{i,k}\mathbf{P}_{i,k}\mathbf{P}_{i,k}^H\mathbf{H}_{i,k}^H\right)\mathbf{N}_k^{-1}\right), \quad (2)$$

where $\mathbf{N}_k\triangleq\sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\mathbf{P}_{j,l}\mathbf{P}_{j,l}^H\mathbf{H}_{j,k}^H+\sigma_k^2\mathbf{I}$ represents the interference-plus-noise covariance matrix. Table I lists all key symbols used in this paper.

Remark 1: We note here that NCJT represents a typical multi-AP JT scheme, which is network agnostic and applicable to a variety of studied network scenarios. One typical instantiation is the NCJT C-RAN network [4], wherein geographically distributed remote radio units (RRUs) are coordinated by the baseband unit (BBU) pool, in an NCJT manner, to jointly serve UEs. Besides, APs in other cooperating MIMO networks, such as dense small cell networks [10] and cell-free networks [11], can also cooperate in the NCJT manner.

B. Problem Formulation

In light of the fact that stream allocation should adapt to the channel quality in practice, our goal is jointly optimizing downlink beamforming and stream allocation to maximize the system WSR, subject to both the individual power budget at

[1]We assume that $\mathbf{H}_{i,k}$ is known at AP i, which can be acquired by existing channel estimation algorithms [31] and reasonable pilot allocation strategies [32]. Moreover, the proposed algorithms are designed within the WMMSE framework, which has been demonstrated to be effective in scenarios with channel estimation errors [16], [17], [18].

[2]The data rate in (2) is achievable assuming the effective channel $\mathbf{H}_{i,k}\mathbf{P}_{i,k}$ is known at UE k and perfect successive interference cancellation (SIC) is conducted. The former assumption can be realized by downlink piloting and the latter can be realized by utilizing strong error correction codes [33].

each AP and the data stream limit at each UE. Accordingly, the optimization problem can be formulated as

$$\max_{\substack{\{\mathbf{P}_{i,k}\},\\ \{D_{i,k}\}}} \sum_{k\in\mathcal{K}} \alpha_k \log\det\left(\mathbf{I}+\left(\sum_{i\in\mathcal{I}_k}\mathbf{H}_{i,k}\mathbf{P}_{i,k}\mathbf{P}_{i,k}^H\mathbf{H}_{i,k}^H\right)\mathbf{N}_k^{-1}\right) \tag{3a}$$

$$\text{s.t.} \quad \sum_{k\in\mathcal{K}_i} \mathrm{Tr}\left(\mathbf{P}_{i,k}\mathbf{P}_{i,k}^H\right) \le P_{\max,i}, \ \forall i, \tag{3b}$$

$$\sum_{i\in\mathcal{I}_k} D_{i,k} \le N_k, \ \forall k, \tag{3c}$$

$$D_{i,k} \in \mathbb{N}, \ \forall k, \ \forall i, \tag{3d}$$

where $\alpha_k > 0$ is the weight of the data rate of UE k, indicating the priority of this UE [4], [10], $P_{\max,i}$ is the power budget of AP i. Here, $D_{i,k}$ is the number of streams sent from AP i to UE k, corresponding to the number of columns in $\mathbf{P}_{i,k}$.

Note that existing studies on the joint optimization of beamforming and DSA mostly address the single-AP network. These studies employ either heuristic approaches that compromise WSR [26] or greedy-based methods [27], [28], [29] that incur high computational complexity. Moreover, while some works consider the multi-AP network, they only focus on AP-UE pairing, and neglect the crucial aspect of DSA. Specifically, the AP-UE pairing problem, such as clustering [9], user scheduling [11], user pairing [12], and user association [13], is indeed a degenerate case of the DSA problem (3), in the sense that whether AP i is selected as the serving unit for UE k corresponds to whether $D_{i,k}$ is larger than zero. As a result, our proposed method is also capable of addressing the AP-UE pairing problem, as elaborated in Section IV.

Obviously, the joint beamforming and DSA optimization problem (3) is mixed-integer programming and nonconvex with tightly coupled decision variables, which is known to be NP-hard. Specifically, the challenges associated with solving problem (3) are two-fold. First, each beamforming matrix $\mathbf{P}_{i,k}$ is tightly coupled with the integer number of data streams $D_{i,k}$, as it determines the dimension of $\mathbf{P}_{i,k}$. Second, even if the stream allocation variables are fixed, simplifying (3) to a pure beamforming optimization problem, the reduced problem remains non-convex and NP-hard.

Given the aforementioned challenges of addressing problem (3), we first consider the pure beamforming optimization problem by fixing the stream allocation variables $\{D_{i,k}\}$ in Section III, and present efficient algorithms for solving this reduced problem. Based on these methods, we then propose a joint beamforming and DSA algorithm in Section IV to address the original problem (3) effectively.

III. Beamforming Design With Fixed Streams

With fixed $\{D_{i,k}\}$, the joint beamforming and DSA problem (3) reduces to

$$\max_{\{\mathbf{P}_{i,k}\}} \sum_{k\in\mathcal{K}} \alpha_k \log\det\left(\mathbf{I}+\left(\sum_{i\in\mathcal{I}_k}\mathbf{H}_{i,k}\mathbf{P}_{i,k}\mathbf{P}_{i,k}^H\mathbf{H}_{i,k}^H\right)\mathbf{N}_k^{-1}\right)$$
$$\text{s.t.} \quad \text{(3b)}. \tag{4}$$

Although various approaches have been proposed to solve this problem [4], [10], [11], [12], these methods are often plagued by the computational complexity issue. The authors in [4], [12] proved that the SCA-based beamforming algorithm achieves a computational complexity of $\mathcal{O}\left(\left(\sum_{i\in\mathcal{I}} M_i \sum_{k\in\mathcal{K}_i} D_{i,k}\right)^3\right)$ (recalling that the number of transmit antennas M_i can be hundreds in massive MIMO [14]). The authors in [10] proposed an approach that integrates InAP and ADMM with a lower complexity in the order of $\mathcal{O}\left(\left(\sum_{i\in\mathcal{I}} M_i\right)^3\right)$. Furthermore, the work [11] proposed an FP-based NCJT beamforming algorithm, which reduces the computational complexity to $\mathcal{O}\left(\sum_{i\in\mathcal{I}} M_i^3\right)$. Moreover, [11], along with [4], [10], [12], only considered networks where each UE has a single antenna.

Note that the computational complexity of these beamforming algorithms scales cubically with the number of AP antennas [11], or even with the total number of antennas across all APs [4], [10], [12], resulting in substantial computational overhead, particularly in massive MIMO. To tackle the complexity issue, we propose a WMMSE-type algorithm, named distributed RWMMSE, which significantly reduces the computational complexity to a linear relationship with M_i, without incurring any WSR loss.

To the best of our knowledge, although an efficient WMMSE-type algorithm has been proposed for CJT [16], none of the NCJT beamforming problems have been solved via the WMMSE framework. In comparison with NCJT, the primary distinction of CJT is that serving APs $\mathcal{I}_k$ in CJT transmit the *same* signal $\mathbf{s}_k$ to UE k, i.e., $\mathbf{s}_k = \mathbf{s}_{i,k}, \forall i \in \mathcal{I}_k$. The achievable data rate of UE k in CJT can be computed as

$$R_{k,\mathrm{CJT}} = \log\det\left(\mathbf{I}+\left(\sum_{i\in\mathcal{I}_k}\mathbf{H}_{i,k}\mathbf{P}_{i,k}\right)\left(\sum_{i\in\mathcal{I}_k}\mathbf{H}_{i,k}\mathbf{P}_{i,k}\right)^H\mathbf{N}_{k,\mathrm{CJT}}^{-1}\right), \tag{5}$$

where the interference-plus-noise term $\mathbf{N}_{k,\mathrm{CJT}} \triangleq \sigma_k^2\mathbf{I} + \left(\sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\mathbf{P}_{j,l}\right)\left(\sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\mathbf{P}_{j,l}\right)^H$. The rate expression R_k in (2) is largely different from $R_{k,\mathrm{CJT}}$, because the signal covariance matrix in (2) appears to be not in the gram matrix form (i.e., $\mathbf{G}\mathbf{G}^H$). This creates obstacles for direct applications of the classic WMMSE techniques [16] to NCJT beamforming [4], [10].

In order to present the distributed RWMMSE algorithm, this section first extends the classic (but centralized) WMMSE algorithm [16] to solve problem (4), which demonstrates the effectiveness of the WMMSE framework in NCJT networks. Based on this, we further introduce the distributed RWMMSE algorithm that has much lower computational and interaction costs between APs and the CPU, but compromises no WSR.

A. Centralized WMMSE Beamforming Algorithm

In essence, the classic WMMSE algorithm equivalently transforms the original non-convex Shannon capacity maximization problem into a more tractable weighted sum mean square error (MSE) minimization problem by introducing auxiliary variables, which is then solved by the block coordinate descent (BCD) method with closed-form updates[3] [16], [35]. This equivalent WMMSE transformation is summarized in the following lemma.

Lemma 1 [35, Lemma 4.1]: For any $\mathbf{A} \in \mathbb{C}^{n\times p}$, $\mathbf{B} \in \mathbb{C}^{p\times m}$ and $\mathbf{N} \succ \mathbf{0} \in \mathbb{C}^{n\times n}$, the following transformation holds:

$$\begin{aligned}&\log\det(\mathbf{I}+\mathbf{A}\mathbf{B}\mathbf{B}^H\mathbf{A}^H\mathbf{N}^{-1})\\&=\max_{\mathbf{C}\succ\mathbf{0},\mathbf{Y}}\log\det(\mathbf{C})-\mathrm{Tr}(\mathbf{C}\mathbf{E}(\mathbf{Y},\mathbf{B}))+m,\end{aligned}\tag{6}$$

where $\mathbf{C} \in \mathbb{C}^{m\times m}$ and $\mathbf{Y} \in \mathbb{C}^{n\times m}$ are auxiliary variables, and the MSE matrix $\mathbf{E} \in \mathbb{C}^{m\times m}$ is defined as

$$\mathbf{E}(\mathbf{Y},\mathbf{B}) \triangleq \left(\mathbf{I}-\mathbf{Y}^H\mathbf{A}\mathbf{B}\right)\left(\mathbf{I}-\mathbf{Y}^H\mathbf{A}\mathbf{B}\right)^H+\mathbf{Y}^H\mathbf{N}\mathbf{Y}. \tag{7}$$

The optimal $\mathbf{Y}^\star$ and $\mathbf{C}^\star$ for (6) are given by

$$\mathbf{Y}^\star=\left(\mathrm{N}+\mathrm{ABB}^H\ \mathrm{A}^H\right)^{-1}\mathrm{AB}, \tag{8a}$$

$$\mathbf{C}^\star=\left(\mathbf{I}-(\mathbf{Y}^\star)^H\mathbf{A}\mathbf{B}\right)^{-1}. \tag{8b}$$

In order to apply Lemma 1 to the NCJT beamforming problem (4), we first define the concatenation of the channel matrices and beamformers from APs $\mathcal{I}_k$ to UE k as

$$\mathbf{H}_k \triangleq [\mathbf{H}_{i_1,k},\mathbf{H}_{i_2,k},\ldots,\mathbf{H}_{i_{|\mathcal{I}_k|},k}] \in \mathbb{C}^{N_k\times M_{(k)}}, \tag{9}$$

$$\mathbf{P}_k \triangleq \mathrm{blkdiag}(\mathbf{P}_{i_1,k},\mathbf{P}_{i_2,k},\ldots,\mathbf{P}_{i_{|\mathcal{I}_k|},k}) \in \mathbb{C}^{M_{(k)}\times D_{(k)}}, \tag{10}$$

where $M_{(k)} \triangleq \sum_{i\in\mathcal{I}_k} M_i$, $D_{(k)} \triangleq \sum_{i\in\mathcal{I}_k} D_{i,k}$, and $|\mathcal{I}_k|$ is the number of APs in $\mathcal{I}_k$. Based on (9) and (10), problem (4) can be recast as

$$\begin{aligned}&\max_{\{\mathbf{P}_{i,k}\}}\sum_{k\in\mathcal{K}}\alpha_k\log\det\left(\mathbf{I}+\mathbf{H}_k\mathbf{P}_k\mathbf{P}_k^H\mathbf{H}_k^H\mathbf{N}_k^{-1}\right)\\&\ \text{s. t. (3b)}.\end{aligned}\tag{11}$$

Now, the structure of the rate expression in problem (11) precisely matches the WMMSE transformation in Lemma 1, allowing problem (11) to be equivalently transformed to

$$\begin{aligned}&\max_{\substack{\{\mathbf{P}_{i,k}\},\{\mathbf{U}_k\}\\\{\mathbf{W}_k\succ\mathbf{0}\}}}\sum_{k\in\mathcal{K}}\alpha_k\left(\log\det(\mathbf{W}_k)-\mathrm{Tr}\left(\mathbf{W}_k\mathbf{E}_k(\mathbf{U}_k,\mathcal{P})\right)\right)\\&\ \text{s. t. (3b)},\end{aligned}\tag{12}$$

where $\mathcal{P} \triangleq \{\mathbf{P}_{i,k}\}$, $\mathbf{U}_k \in \mathbb{C}^{N_k\times D_{(k)}}$, $\mathbf{W}_k \succ \mathbf{0} \in \mathbb{C}^{D_{(k)}\times D_{(k)}}$ are the auxiliary variables, and the MSE matrix $\mathbf{E}_k \in \mathbb{C}^{D_{(k)}\times D_{(k)}}$ is defined as

$$\begin{aligned}\mathbf{E}_k(\mathbf{U}_k,\mathcal{P}) \triangleq &\left(\mathbf{I}-\mathbf{U}_k^H\mathbf{H}_k\mathbf{P}_k\right)\left(\mathbf{I}-\mathbf{U}_k^H\mathbf{H}_k\mathbf{P}_k\right)^H\\&+\mathbf{U}_k^H\mathbf{N}_k\mathbf{U}_k.\end{aligned}\tag{13}$$

Note that problem (12) is convex with respect to (w.r.t.) $\{\mathbf{U}_k\}$, $\{\mathbf{W}_k\}$ and $\{\mathbf{P}_{i,k}\}$, respectively. According to (8), the optimal $\{\mathbf{U}_k^\star\}$ and $\{\mathbf{W}_k^\star\}$ with fixed $\{\mathbf{P}_{i,k}\}$ are given by, respectively,

$$\mathbf{U}_k^\star=\left(\mathbf{N}_k+\mathbf{H}_k\mathbf{P}_k\mathbf{P}_k^H\mathbf{H}_k^H\right)^{-1}\mathbf{H}_k\mathbf{P}_k,\ \forall k, \tag{14}$$

$$\mathbf{W}_k^\star=(\mathbf{I}-(\mathbf{U}_k^\star)^H\mathbf{H}_k\mathbf{P}_k)^{-1},\ \forall k. \tag{15}$$

[3] The BCD method is widely used for minimizing continuous functions with multiple block variables. During each iteration of this method, one block of variables is optimized while the remaining blocks are held fixed [34].

Algorithm 1 Centralized WMMSE Beamforming Algorithm

Interaction: Each AP i sends $\{\mathbf{H}_{i,k} : k \in \mathcal{K}\}$ to the CPU pool;
1: Initialize $\{\mathbf{P}_{i,k}\}$;
2: **Repeat**
3: Update $\{\mathbf{U}_k^\star\}$ via (14);
4: Update $\{\mathbf{W}_k^\star\}$ via (15);
5: Update $\{\mathbf{P}_{i,k}^\star\}$ via (17);
6: **Until** convergence
Interaction: Send $\{\mathbf{P}_{i,k}^\star : k \in \mathcal{K}_i\}$ back to each AP i.

By fixing $\{\mathbf{U}_k\}$ and $\{\mathbf{W}_k\}$ in (12), the optimization problem w.r.t. $\{\mathbf{P}_{i,k}\}$ can be written as

$$\begin{aligned}\min_{\{\mathbf{P}_{i,k}\}}&\sum_{k\in\mathcal{K}}\alpha_k\sum_{i\in\mathcal{I}_k}\mathrm{Tr}\left(\mathbf{P}_{i,k}^H\mathbf{H}_{i,k}^H\mathbf{A}_k\mathbf{H}_{i,k}\mathbf{P}_{i,k}\right)\\&-2\sum_{k\in\mathcal{K}}\alpha_k\sum_{i\in\mathcal{I}_k}\mathrm{Re}\left\{\mathrm{Tr}\left(\mathbf{\Xi}_{i,k}\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_{i,k}\mathbf{P}_{i,k}\right)\right\}\\&+\sum_{k\in\mathcal{K}}\alpha_k\mathrm{Tr}\left(\mathbf{A}_k\sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\mathbf{P}_{j,l}\mathbf{P}_{j,l}^H\mathbf{H}_{j,k}^H\right)\\\text{s. t.}&\text{(3b)},\end{aligned}\tag{16}$$

where $\mathbf{A}_k \triangleq \mathbf{U}_k\mathbf{W}_k\mathbf{U}_k^H$, $\mathbf{\Xi}_{i,k} \in \mathbb{B}^{D_{i,k}\times D_{(k)}}$ is defined such that its sub-matrix generated from the $((i-1)\times D_{i,k}+1)$-th column to the $(i\times D_{i,k})$-th column being $\mathbf{I}_{D_{i,k}}$ and the rest elements being zero. Here, $\mathbf{\Xi}_{i,k}$ is specifically designed to extract the $D_{i,k}$ rows that are related to $\mathbf{P}_{i,k}$ from $\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_{i,k}\mathbf{P}_{i,k}$. Since problem (16) is convex, the optimal $\{\mathbf{P}_{i,k}^\star\}$ can be computed by the Lagrangian multiplier method, given by

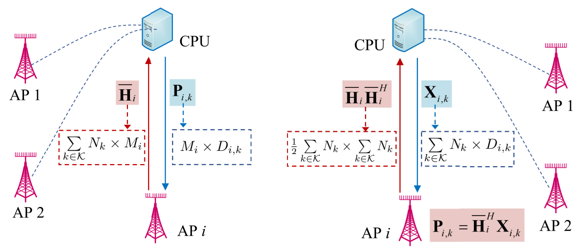
$$\mathbf{P}_{i,k}^\star=\left(\sum_{l\in\mathcal{K}}\alpha_l\mathbf{H}_{i,l}^H\mathbf{A}_l\mathbf{H}_{i,l}+\mu_i\mathbf{I}\right)^{-1}\alpha_k\mathbf{H}_{i,k}^H\mathbf{U}_k\mathbf{W}_k\mathbf{\Xi}_{i,k}^H, \tag{17}$$

where $\mu_i \geq 0$ is the Lagrangian multiplier associated with the maximum transmit power constraint (3b), which can be efficiently solved using one-dimensional search techniques [16], e.g., bisection method. Combining (14), (15), and (17), we summarize the centralized WMMSE algorithm in Algorithm 1.

The computational complexity of WMMSE is primarily determined by the inverse of $\sum_{l\in\mathcal{K}}\alpha_l\mathbf{H}_{i,l}^H\mathbf{A}_l\mathbf{H}_{i,l}+\mu_i\mathbf{I}$, which is given as $\mathcal{O}(M_i^3)$. This cubic computational complexity can result in significant computational cost, particularly when APs are equipped with massive antennas. Additionally, the WMMSE algorithm is centralized, which requires the transmission of raw channel matrices $\{\mathbf{H}_{i,k} : k \in \mathcal{K}\}$ from each AP to the CPU, as shown in Fig. 2(a), introducing substantial interaction overhead. The high computational complexity and interaction cost motivate us to explore more efficient beamforming algorithms in the next subsection.

B. Distributed RWMMSE Beamforming Algorithm

In the sequel, we first study the low-dimensional subspace property of $\mathbf{P}_{i,k}^\star$ in (17). This intrinsic property will subsequently be exploited to design the distributed RWMMSE algorithm that can reduce both the computational complexity



(a) WMMSE (b) RWMMSE

Fig. 2. An exemplary illustration of the interaction between the APs and the CPU in the WMMSE and RWMMSE algorithms, where $M_i \geq N_{\text{total}}$.

and the interaction cost without any WSR loss compared to the original centralized WMMSE algorithm.

1) WMMSE Reshaping: For ease of expression, we concat the channel matrices between AP i and all the UEs as

$$\bar{\mathbf{H}}_i \triangleq \left[\mathbf{H}_{i,1}^H, \ldots, \mathbf{H}_{i,K}^H\right]^H \in \mathbb{C}^{N_{\text{total}} \times M_i}, \tag{18}$$

where $N_{\text{total}} \triangleq \sum_{k\in\mathcal{K}} N_k$. Then, the update of $\mathbf{P}_{i,k}^\star$ given in (17) can be compactly rewritten as

$$\mathbf{P}_{i,k}^\star = \left(\bar{\mathbf{H}}_i^H \mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i + \mu_i \mathbf{I}\right)^{-1} \alpha_k \mathbf{H}_{i,k}^H \mathbf{U}_k \mathbf{W}_k \mathbf{\Xi}_{i,k}^H, \tag{19}$$

with $\mathbf{U} \triangleq \operatorname{blkdiag}(\mathbf{U}_1, \ldots, \mathbf{U}_K)$ and $\mathbf{W} \triangleq \operatorname{blkdiag}(\alpha_1 \mathbf{W}_1, \ldots, \alpha_K \mathbf{W}_K)$.

Since $\mathbf{H}_{i,k}^H$ in (19) is exactly the sub-columns of $\bar{\mathbf{H}}_i^H$, we can express it as $\mathbf{H}_{i,k}^H = \bar{\mathbf{H}}_i^H \mathbf{T}_{i,k}$ with $\mathbf{T}_{i,k} \triangleq \left[\mathbf{0}_{N_k \times \sum_{l=1}^{k-1} N_l}, \mathbf{I}_{N_k}, \mathbf{0}_{N_k \times \sum_{l=k+1}^{K} N_l}\right]^H$. Then, by applying the Woodbury identity, i.e., $(\mathbf{I} + \mathbf{A}\mathbf{B})^{-1} \mathbf{A} = \mathbf{A}(\mathbf{I} + \mathbf{B}\mathbf{A})^{-1}$, we establish a low-dimensional subspace structure for beamforming in the following theorem.[4]

Theorem 1: Any stationary point $\mathbf{P}_{i,k}^\star$ of problem (11) must lie in the column space of $\bar{\mathbf{H}}_i^H$, i.e.,

$$\mathbf{P}_{i,k}^\star = \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}^\star, \tag{20}$$

where

$$\mathbf{X}_{i,k}^\star = \alpha_k \left(\mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H + \mu_i \mathbf{I}\right)^{-1} \mathbf{T}_{i,k} \mathbf{U}_k \mathbf{W}_k \mathbf{\Xi}_{i,k}^H \tag{21}$$

is a low-dimensional substitution of $\mathbf{P}_{i,k}^\star$ with size $N_{\text{total}} \times D_{i,k}$.

Proof: See Appendix A. ∎

Remark 2: The low-dimensionality of $\mathbf{X}_{i,k}^\star$ in Theorem 1 implies $M_i \geq N_{\text{total}}$. Indeed, it is common practice for $M_i \geq N_{\text{total}}$ in MU-MIMO network configurations, as N_{total} actually indicates the total receive antennas of all UEs sharing the same time-frequency resource, with transmissions on different time-frequency resources typically being non-interfering [12]. Moreover, practical implementations of MU-MIMO must address multi-user interference (MUI) management among UEs sharing the same time-frequency resource by leveraging spatial diversity. However, due to the inherent limitations of MUI suppression capabilities, N_{total} cannot be excessively large, even if there may be dozens or even hundreds of UEs across the entire frequency band. For instance, the 3GPP technical specification (TS) 38.211 specifies only 8 independent data streams for each AP [37]. Thus, we focus on the case when $M_i \geq N_{\text{total}}$ to illustrate the proposed algorithms, unless otherwise stated. We will explain how to apply the proposed algorithms to the case when $M_i < N_{\text{total}}$ at the end of this subsection.

Theorem 1 implies that the optimal $\mathbf{P}_{i,k}^\star$ lies in the column space of $\bar{\mathbf{H}}_i^H$. As such, we can also obtain the optimal $\mathbf{X}_{i,k}^\star$ by substituting $\mathbf{P}_{i,k} = \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}$ into problem (4) and solving the resultant problem using the WMMSE framework. Then a natural question arises: Does the optimal $\mathbf{X}_{i,k}^\star$ obtained in this way coincide with the result given in (21)?

The answer is affirmative, as explained below. By substituting $\mathbf{P}_{i,k} = \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}$ into problem (4) and applying the WMMSE framework (or directly substituting $\mathbf{P}_{i,k} = \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}$ into problem (16)), the optimization problem for $\{\mathbf{X}_{i,k}\}$ is given by

$$\begin{aligned}
\min_{\{\mathbf{X}_{i,k}\}} \; & \sum_{k\in\mathcal{K}} \alpha_k \sum_{i\in\mathcal{I}_k} \operatorname{Tr}\left(\mathbf{X}_{i,k}^H \bar{\mathbf{H}}_i \mathbf{H}_{i,k}^H \mathbf{A}_k \mathbf{H}_{i,k} \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}\right) \\
& - 2\sum_{k\in\mathcal{K}} \alpha_k \sum_{i\in\mathcal{I}_k} \operatorname{Re}\left\{\operatorname{Tr}\left(\mathbf{\Xi}_{i,k} \mathbf{W}_k \mathbf{U}_k^H \mathbf{H}_{i,k} \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}\right)\right\} \\
& + \sum_{k\in\mathcal{K}} \alpha_k \operatorname{Tr}\left(\mathbf{A}_k \sum_{j\in\mathcal{I}_l} \sum_{l\in\mathcal{K}_{-k}} \mathbf{H}_{j,k} \bar{\mathbf{H}}_j^H \mathbf{X}_{j,l} \mathbf{X}_{j,l}^H \bar{\mathbf{H}}_j \mathbf{H}_{j,k}^H\right) \\
\text{s.t.} \; & \sum_{k\in\mathcal{K}_i} \operatorname{Tr}\left(\bar{\mathbf{H}}_i^H \mathbf{X}_{i,k} \mathbf{X}_{i,k}^H \bar{\mathbf{H}}_i\right) \leq P_{\max,i}, \; \forall i.
\end{aligned} \tag{22}$$

Using the Lagrangian multiplier method, the optimal $\mathbf{X}_{i,k}^\ddagger$ is given by

$$\begin{aligned}
\mathbf{X}_{i,k}^\ddagger = & \left(\sum_{l\in\mathcal{K}} \alpha_l \bar{\mathbf{H}}_i \mathbf{H}_{i,l}^H \mathbf{A}_l \mathbf{H}_{i,l} \bar{\mathbf{H}}_i^H + \lambda_i \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H\right)^{-1} \\
& \times \alpha_k \bar{\mathbf{H}}_i \mathbf{H}_{i,k}^H \mathbf{U}_k \mathbf{W}_k \mathbf{\Xi}_{i,k}^H,
\end{aligned} \tag{23}$$

where $\lambda_i \geq 0$ is the Lagrangian multiplier. Although the two computations in (21) and (23) seem quite different, the following proposition confirms that they are exactly the same.

Proposition 1: Equations (21) and (23) are equivalent to each other.

Proof: See Appendix B. ∎

Theorem 1 and Proposition 1 demonstrate that we can utilize (20) to determine $\mathbf{P}_{i,k}^\star$, where $\mathbf{X}_{i,k}^\star$ can be computed via either (21) or (23). Both of these computations have a complexity that is cubic in N_{total}, since they are dominated by the inversion of the matrices $\mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H + \mu_i \mathbf{I}$ in (21) or $\sum_{l\in\mathcal{K}} \alpha_l \bar{\mathbf{H}}_i \mathbf{H}_{i,l}^H \mathbf{A}_l \mathbf{H}_{i,l} \bar{\mathbf{H}}_i^H + \lambda_i \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H$ in (23) with size $N_{\text{total}} \times N_{\text{total}}$. This represents a reduction in complexity compared to the original complexity of $\mathcal{O}(M_i^3)$ when computing $\mathbf{P}_{i,k}^\star$ using (17). Furthermore, we reshape the update of $\mathbf{X}_{i,k}$ to attain an even lower computational complexity.

Corollary 1: The low-dimensional substitution $\mathbf{X}_{i,k}$ in (21) and (23) can be rewritten as follows:

$$\mathbf{X}_{i,k}^\star = \mathbf{U}\mathbf{W}\left(\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H \mathbf{U}\mathbf{W} + \mu_i \mathbf{I}\right)^{-1} \mathbf{K}_{i,k} \mathbf{\Xi}_{i,k}^H, \tag{24}$$

where $\mathbf{K}_{i,k} \triangleq \left[\mathbf{0}_{D_{(k)} \times \sum_{l=1}^{k-1} D_{(l)}}, \mathbf{I}_{D_{(k)}}, \mathbf{0}_{D_{(k)} \times \sum_{l=k+1}^{K} D_{(l)}}\right]^H$.

[4] A low-dimensional subspace structure similar to Theorem 1 was first presented in our previous work [36, Proposition 2]. However, [36] is restricted to single-AP networks, whereas Theorem 1 applies to the more general multi-AP scenario. Furthermore, the proof in [36] relies on subspace theory, while this paper offers a more concise proof utilizing the Woodbury identity.

Corollary 1 can be easily proved using the Woodbury identity and the equality $\alpha_k \mathbf{T}_{i,k}\mathbf{U}_k\mathbf{W}_k = \mathbf{U}\mathbf{W}\mathbf{K}_{i,k}$, thus is omitted. Using (24), the complexity of computing $\mathbf{X}_{i,k}$ reduces to $\mathcal{O}(N_{\text{total}}^2 D_{\text{total}})$ with $D_{\text{total}} \triangleq \sum_{k\in\mathcal{K}} D_{(k)}$, which is smaller than the $\mathcal{O}(N_{\text{total}}^3)$ required for (21) and (23).

2) Distributed RWMMSE Algorithm: Based on the above analysis, now we are ready to propose a more efficient WMMSE-type algorithm with reduced complexity and interaction cost. The main idea is to replace the update of $\{\mathbf{P}_{i,k}\}$ with its low-dimensional substitution $\{\mathbf{X}_{i,k}\}$, which requires only $\{\mathbf{H}_{i,k}\mathbf{H}_{i,k}^H\}$ rather than the raw channel matrices $\{\mathbf{H}_{i,k}\}$ as in traditional WMMSE.

Following (20) and recalling the definition of $\mathbf{P}_k$ in (10), we have

$$\mathbf{P}_k = \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \tilde{\mathbf{X}}_{\mathcal{I}_k}, \tag{25}$$

where $\tilde{\mathbf{H}}_{\mathcal{I}_k} \triangleq \text{blkdiag}(\bar{\mathbf{H}}_{i_1}, \ldots, \bar{\mathbf{H}}_{i_{|\mathcal{I}_k|}}) \in \mathbb{C}^{|\mathcal{I}_k|N_{\text{total}} \times M_{(k)}}$ and $\tilde{\mathbf{X}}_{\mathcal{I}_k} \triangleq \text{blkdiag}(\mathbf{X}_{i_1,k}, \ldots, \mathbf{X}_{i_{|\mathcal{I}_k|},k}) \in \mathbb{C}^{|\mathcal{I}_k|N_{\text{total}} \times D_{(k)}}$ are the concatenations of the channel matrices $\{\bar{\mathbf{H}}_i : i \in \mathcal{I}_k\}$ and the low-dimensional matrices $\{\mathbf{X}_{i,k} : i \in \mathcal{I}_k\}$, respectively.

By substituting (25) into (14) and (15), the new updates of $\{\mathbf{U}_k^\star\}$ and $\{\mathbf{W}_k^\star\}$ are given by, respectively,

$$\mathbf{U}_k^\star = \left(\tilde{\mathbf{N}}_k + \mathbf{H}_k \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \tilde{\mathbf{X}}_{\mathcal{I}_k} \tilde{\mathbf{X}}_{\mathcal{I}_k}^H \tilde{\mathbf{H}}_{\mathcal{I}_k} \mathbf{H}_k^H\right)^{-1} \times \mathbf{H}_k \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \tilde{\mathbf{X}}_{\mathcal{I}_k}, \ \forall k, \tag{26}$$

$$\mathbf{W}_k^\star = \left(\mathbf{I} - (\mathbf{U}_k^\star)^H \mathbf{H}_k \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \tilde{\mathbf{X}}_{\mathcal{I}_k}\right)^{-1}, \ \forall k, \tag{27}$$

where

$$\tilde{\mathbf{N}}_k \triangleq \sum_{l\in\mathcal{K}_{-k}} \sum_{j\in\mathcal{I}_l} \mathbf{H}_{j,k}\bar{\mathbf{H}}_j^H \mathbf{X}_{j,l}\mathbf{X}_{j,l}^H \bar{\mathbf{H}}_j \mathbf{H}_{j,k}^H + \sigma_k^2 \mathbf{I}. \tag{28}$$

Note that $\mathbf{H}_k\tilde{\mathbf{H}}_{\mathcal{I}_k}^H$ in the new updates above is the concatenation of $\{\mathbf{H}_{i,k}\bar{\mathbf{H}}_{i,l}^H : i \in \mathcal{I}_k, l \in \mathcal{K}\}$, which is a sub-matrix of $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H \in \mathbb{C}^{N_{\text{total}} \times N_{\text{total}}}$. Similarly, the term $\mathbf{H}_{j,k}\bar{\mathbf{H}}_j^H$ in (28) is a sub-matrix of $\bar{\mathbf{H}}_j\bar{\mathbf{H}}_j^H$. Consequently, combining the updates of $\{\mathbf{U}_k\}$, $\{\mathbf{W}_k\}$ and $\{\mathbf{X}_{i,k}\}$ in (26), (27) and (24), the distributed RWMMSE algorithm only requires low-dimensional matrices $\{\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H\}$, rather than the raw channel matrices $\{\bar{\mathbf{H}}_i\}$ needed in the WMMSE algorithm.

Algorithm 2 Distributed RWMMSE Beamforming Algorithm

1: Each AP i calculates $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$;
   Interaction: Each AP i sends $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$ to the CPU pool;
2: Initialize $\{\mathbf{X}_{i,k}\}$;
3: **Repeat**
4: Update $\{\mathbf{U}_k^\star\}$ via (26);
5: Update $\{\mathbf{W}_k^\star\}$ via (27);
6: Update $\{\mathbf{X}_{i,k}^\star\}$ via (24);
7: **Until** convergence
   Interaction: Send $\{\mathbf{X}_{i,k}^\star : k \in \mathcal{K}_i\}$ back to each AP i;

Output: Each AP i calculates $\{\mathbf{P}_{i,k}^\star = \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}^\star : k \in \mathcal{K}_i\}$.

Finally, we summarize the proposed algorithm in Algorithm 2, and describe its implementation in Fig. 2(b). When executing the distributed RWMMSE algorithm, each AP i first transmits $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$ to the CPU, which has smaller size than the raw channel matrix $\bar{\mathbf{H}}_i$ that needs to be interacted in the WMMSE algorithm (see Fig. 2(a)). The CPU then calculates and transmits $\{\mathbf{X}_{i,k}^\star \in \mathbb{C}^{N_{\text{total}} \times D_{i,k}} : k \in \mathcal{K}_i\}$ back to each AP i, rather than the complete beamformers $\{\mathbf{P}_{i,k}^\star : k \in \mathcal{K}_i\}$ in the WMMSE algorithm. Finally, each AP i calculates its beamformer via (20). Note that our RWMMSE algorithm is *distributed* in the sense that it fits into the star decentralized baseband processing (DBP) architecture [38], requiring neither raw CSI interaction nor raw beamformer interaction, and the complete beamforming matrices are obtained *locally*. For simplicity, RWMMSE hereafter refers to distributed RWMMSE.

3) Initialization of RWMMSE: In sequel, we introduce an initialization approach for the RWMMSE algorithm. The computational complexity and interaction cost of the proposed methods will be analyzed in subsection III-C.

A natural question is how to efficiently initialize $\mathbf{X}_{i,k}$, while satisfying the individual power constraints (3b). Recalling that eigen zero-forcing (EZF) is a simple but very effective beamforming method for CJT [39], we here develop a local EZF method for NCJT. Unlike EZF in CJT that requires APs to interact CSI to coherently transmit the same data, the proposed local EZF method does not necessitate data interaction, since each AP sends distinct data streams to the UE in NCJT. Furthermore, this method meets the low-dimensional property in Theorem 1, and thus can be used to efficiently initialize the RWMMSE algorithm.

The local EZF method follows the basic idea behind EZF, i.e., perform singular value decomposition (SVD) on the channel matrix first, and then apply ZF on the effective channel matrix composed of the right singular vectors. Specifically, each AP i first performs SVD on $\{\mathbf{H}_{i,k} : k \in \mathcal{K}_i\}$ to obtain $\mathbf{H}_{i,k} = \mathbf{Q}_{i,k}\mathbf{\Sigma}_{i,k}\mathbf{V}_{i,k}^H$, where $\mathbf{Q}_{i,k} \in \mathbb{C}^{N_k \times N_k}$ and $\mathbf{V}_{i,k} \in \mathbb{C}^{M_i \times N_k}$ are unitary matrices composed of the left and right singular vectors, respectively, and $\mathbf{\Sigma}_{i,k} \in \mathbb{R}^{N_k \times N_k}$ is a diagonal matrix with descending singular values on the diagonal.

The effective channel of AP i, i.e., $\tilde{\mathbf{V}}_i \in \mathbb{C}^{M_i \times \sum_{k\in\mathcal{K}_i} D_{i,k}}$ is then formed by concating the first $D_{i,k}$ columns of each $\mathbf{V}_{i,k}$, for $k \in \mathcal{K}_i$, as

$$\tilde{\mathbf{V}}_i = [\mathbf{V}_{i,k_1}(:, 1{:}D_{i,k_1}), \ldots, \mathbf{V}_{i,k_{|\mathcal{K}_i|}}(:, 1 : D_{i,k_{|\mathcal{K}_i|}})]. \tag{29}$$

Finally, each AP i performs ZF and power scaling on $\bar{\mathbf{V}}_i$ locally, yielding the local EZF beamformer given by

$$\tilde{\mathbf{P}}_i^{\text{local EZF}} = \frac{\left(\tilde{\mathbf{V}}_i^H\right)^\dagger}{\left\|\left(\tilde{\mathbf{V}}_i^H\right)^\dagger\right\|_F} \sqrt{P_{\max,i}} = [\mathbf{P}_{i,k_1}, \mathbf{P}_{i,k_2}, \ldots, \mathbf{P}_{i,k_{|\mathcal{K}_i|}}], \ \forall i. \tag{30}$$

Algorithm 3 Fully Distributed Local EZF Algorithm

1: Perform SVD on $\{\mathbf{H}_{i,k} : k \in \mathcal{K}_i\}$;
2: Concat $\bar{\mathbf{V}}_i$ via (29) as its effective channel;
3: Calculate the local EZF beamformer $\mathbf{P}_i^{\text{local EZF}}$ via (30).

We summarize the local EZF method in Algorithm 3. In contrast to the aforementioned WMMSE-type algorithms, the local EZF beamformer of each AP only involves its local CSI, i.e., $\{\mathbf{H}_{i,k} : k \in \mathcal{K}_i\}$, thereby eliminating the need for CSI interaction between the AP and the CPU pool.

Similar to Theorem 1, we demonstrate in the following proposition that the local EZF beamformer (30) also conforms to a low-dimensional subspace structure.

Proposition 2: The local EZF beamformer $\{\mathbf{P}_{i,k}^{localEZF} : k \in \mathcal{K}_i\}$ lies in the column space of $\bar{\mathbf{H}}_i^H$ in the form of

$$\tilde{\mathbf{P}}_i^{localEZF} = \tilde{\mathbf{H}}_i^H \tilde{\mathbf{X}}_i^{\text{local EZF}}, \tag{31}$$

where $\tilde{\mathbf{H}}_i$ and $\mathbf{X}_i^{local\ EZF}$ are defined as, respectively,

$$\tilde{\mathbf{H}}_i = [\mathbf{H}_{i,k_1}^H, \mathbf{H}_{i,k_2}^H, \ldots, \mathbf{H}_{i,k_{|\mathcal{K}_i|}}^H]^H \in \mathbb{C}^{\sum_{k\in\mathcal{K}_i} N_k \times M_i} \tag{32}$$

with its column space in the column space of $\bar{\mathbf{H}}_i^H$, and

$$\tilde{\mathbf{X}}_i^{localEZF} = \text{blkdiag}\Big(\mathbf{Q}_{i,k_1}(\mathbf{\Sigma}_{i,k_1}^H)^{-1}\mathbf{J}_{i,k_1}, \ldots, \mathbf{Q}_{i,k_{|\mathcal{K}_i|}}(\mathbf{\Sigma}_{i,k_{|\mathcal{K}_i|}}^H)^{-1}\mathbf{J}_{i,k_{|\mathcal{K}_i|}}\Big)\mathbf{\Gamma}_i \tag{33}$$

with $\mathbf{\Gamma}_i \triangleq \frac{(\tilde{\mathbf{V}}_i^H\tilde{\mathbf{V}}_i)^{-1}}{\|(\tilde{\mathbf{V}}_i^H)^\dagger\|_F}\sqrt{P_{\max,i}}$ and $\mathbf{J}_{i,k} \triangleq [\mathbf{I}_{D_{i,k}}, \mathbf{0}]^T \in \mathbb{B}^{N_k \times D_{i,k}}$ being used to extract the first $D_{i,k}$ columns of $\mathbf{Q}_{i,k}(\mathbf{\Sigma}_{i,k}^H)^{-1}$.

Proof: See Appendix C. ■

Proposition 2 reveals that $\{\mathbf{X}_i^{\text{local EZF}}\}$ can act as an initial point for the RWMMSE iteration. Moreover, since $\mathbf{X}_i^{\text{local EZF}}$ and the RWMMSE update $\mathbf{X}_{i,k}$ in (21) have different number of rows, i.e., $\sum_{k\in\mathcal{K}_i} N_k$ and $\sum_{k\in\mathcal{K}} N_k$, respectively, zero elements need to be inserted into $\mathbf{X}_i^{\text{local EZF}}$ at the corresponding positions of $\tilde{\mathbf{H}}_i$ w.r.t. $\bar{\mathbf{H}}_i$, so as to match $\mathbf{X}_{i,k}$'s size.

4) RWMMSE for $M_i < N_{total}$ Scenarios: We now explain how to apply the proposed RWMMSE algorithm to the case when $M_i < N_{\text{total}}$. In this case, the substitution $\mathbf{X}_{i,k}^\star$ in Theorem 1 has a larger row dimension than the raw beamformer $\mathbf{P}_{i,k}^\star$. Hence, we introduce an alternative low-dimensional substitution for $\mathbf{P}_{i,k}^\star$.

Note that the effective dimension of the column space of $\bar{\mathbf{H}}_i^H$ in Theorem 1 is determined by the rank $r_i = \text{rank}(\bar{\mathbf{H}}_i^H)$, which satisfies $r_i \le \min(M_i, N_{\text{total}})$. This rank constraint implies that the substitution matrix $\mathbf{X}_{i,k}^\star$ can be further reduced in size, while preserving the equivalence with Theorem 1. Specifically, consider the reduced SVD of $\bar{\mathbf{H}}_i^H$, i.e., $\bar{\mathbf{H}}_i^H = \bar{\mathbf{Q}}_i\bar{\mathbf{\Sigma}}_i\bar{\mathbf{V}}_i^H$, where $\bar{\mathbf{Q}}_i \in \mathbb{C}^{M_i\times r_i}$ and $\bar{\mathbf{V}}_i \in \mathbb{C}^{N_{\text{total}}\times r_i}$ are semi-unitary matrices spanning the column and row spaces of $\bar{\mathbf{H}}_i^H$, respectively, and $\bar{\mathbf{\Sigma}}_i \in \mathbb{R}^{r_i\times r_i}$ is a diagonal matrix of positive singular values. We can find another low-dimensional substitution for $\mathbf{P}_{i,k}^\star$, as stated in the following corollary.

Corollary 2: Any stationary point $\mathbf{P}_{i,k}^\star$ must lie in the column space of $\bar{\mathbf{Q}}_i$, and can be expressed as

$$\mathbf{P}_{i,k}^\star = \bar{\mathbf{Q}}_i\bar{\mathbf{\Sigma}}_i\breve{\mathbf{X}}_{i,k}^\star, \tag{34}$$

where $\breve{\mathbf{X}}_{i,k}^\star = \bar{\mathbf{V}}_i^H\mathbf{X}_{i,k}^\star \in \mathbb{C}^{r_i\times D_{i,k}}$ is a reduced-dimensional equivalent of $\mathbf{X}_{i,k}^\star$ defined in (21).

Proof: Substituting $\bar{\mathbf{H}}_i^H = \bar{\mathbf{Q}}_i\bar{\mathbf{\Sigma}}_i\bar{\mathbf{V}}_i^H$ into $\mathbf{P}_{i,k}^\star$ in (20) yields:

$$\mathbf{P}_{i,k}^\star = \bar{\mathbf{Q}}_i\bar{\mathbf{\Sigma}}_i\bar{\mathbf{V}}_i^H\mathbf{X}_{i,k}^\star. \tag{35}$$

Since $\bar{\mathbf{V}}_i$ is semi-unitary, we define the low-dimensional substitution $\breve{\mathbf{X}}_{i,k}^\star \triangleq \bar{\mathbf{V}}_i^H\mathbf{X}_{i,k}^\star$, which projects $\mathbf{X}_{i,k}^\star$ onto the row space of $\bar{\mathbf{V}}_i^H$. Thus, $\mathbf{X}_{i,k}^\star$ can be decomposed as

$$\mathbf{X}_{i,k}^\star = \bar{\mathbf{V}}_i\breve{\mathbf{X}}_{i,k}^\star + \mathbf{E}_i^0, \tag{36}$$

where $\mathbf{E}_i^0$ lies in the null space of $\bar{\mathbf{V}}_i^H$. Substituting (36) into (35), the term $\bar{\mathbf{V}}_i^H\mathbf{E}_i^0$ vanishes due to orthogonality, leading to $\mathbf{P}_{i,k}^\star = \bar{\mathbf{Q}}_i\bar{\mathbf{\Sigma}}_i\breve{\mathbf{X}}_{i,k}^\star$. This confirms that $\mathbf{P}_{i,k}^\star$ is confined to the column space of $\bar{\mathbf{Q}}_i$ and completes the proof. ■

Corollary 2 demonstrates that $\breve{\mathbf{X}}_{i,k}^\star$ has a reduced row dimension r_i, which is not greater than that of the raw beamformer $\mathbf{P}_{i,k}^\star$. Therefore, $\breve{\mathbf{X}}_{i,k}^\star$ can serve as a low-dimensional substitution for $\mathbf{P}_{i,k}^\star$. Moreover, (35) and (36) establish the equivalence between Theorem 1 and Corollary 2, in the sense that $\mathbf{X}_{i,k}^\star$ and $\breve{\mathbf{X}}_{i,k}^\star$ correspond to the same beamformer $\mathbf{P}_{i,k}^\star$, and thus yield the same WSR.

Based on Corollary 2, we can alternately update $\{\breve{\mathbf{X}}_{i,k}\}$ and the auxiliary matrices $\{\mathbf{U}_k\}$ and $\{\mathbf{W}_k\}$ to design the beamforming matrices. Specifically, the updates of $\{\mathbf{U}_k\}$ and $\{\mathbf{W}_k\}$ are the same as (26) and (27) by substituting $\mathbf{X}_{i,k}^\star$ with $\bar{\mathbf{V}}_i\breve{\mathbf{X}}_{i,k}^\star$, and it is not difficult to verify that $\breve{\mathbf{X}}_{i,k}^\star$ can be updated by

$$\begin{aligned}\breve{\mathbf{X}}_{i,k}^\star &= \left(\sum_{l\in\mathcal{K}}\alpha_l\bar{\mathbf{\Sigma}}_i^2\bar{\mathbf{V}}_i^H\mathbf{T}_{i,l}\mathbf{A}_l\mathbf{T}_{i,l}^H\bar{\mathbf{V}}_i\bar{\mathbf{\Sigma}}_i^2 + \lambda_i\bar{\mathbf{\Sigma}}_i^2\right)^{-1} \\ &\quad\times \alpha_k\bar{\mathbf{\Sigma}}_i^2\bar{\mathbf{V}}_i^H\mathbf{T}_{i,k}\mathbf{U}_k\mathbf{W}_k\mathbf{\Xi}_{i,k}^H, \\ &= \bar{\mathbf{V}}_i^H\mathbf{U}\mathbf{W}\left(\mathbf{U}^H\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H\mathbf{U}\mathbf{W} + \mu_i\mathbf{I}\right)^{-1}\mathbf{K}_{i,k}\mathbf{\Xi}_{i,k}^H.\end{aligned} \tag{37}$$

Furthermore, since it is common practice for the channel matrix $\bar{\mathbf{H}}_i^H$ to be almost surely full-rank, we have $r_i = \min(M_i, N_{\text{total}})$ with probability 1. As a result, when $M_i \ge N_{\text{total}}$, the size of $\breve{\mathbf{X}}_{i,k}^\star$ is $M_i \times D_{i,k}$, which is the same as that of $\mathbf{X}_{i,k}^\star$. In this case, using either $\mathbf{X}_{i,k}^\star$ or $\breve{\mathbf{X}}_{i,k}^\star$ results in the same interaction cost, computational complexity per iteration, and WSR. However, we still choose $\mathbf{X}_{i,k}^\star$ as the low-dimensional substitution for $\mathbf{P}_{i,k}^\star$ when $M_i \ge N_{\text{total}}$, as the computing $\breve{\mathbf{X}}_{i,k}^\star$ involves additional SVD computation of $\{\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H\}$ at the CPU to obtain $\{\bar{\mathbf{V}}_i\}$ and $\{\bar{\mathbf{\Sigma}}_i\}$. In contrast, when $M_i < N_{\text{total}}$, the update of $\breve{\mathbf{X}}_{i,k}^\star$ via (37) yields a lower computational complexity than the update of $\mathbf{X}_{i,k}^\star$ via (24). Therefore, when $M_i < N_{\text{total}}$, we use $\breve{\mathbf{X}}_{i,k}^\star$ as the low-dimensional substitution for $\mathbf{P}_{i,k}^\star$, and replace step 6 in Algorithm 2 with updating $\breve{\mathbf{X}}_{i,k}^\star$ via (37).

C. Interaction Cost and Computational Complexity Analysis

In this subsection, we analyze the interaction cost and computational complexity of the proposed algorithms, which are summarized in Table II.

1) Interaction Cost: When transmitting a matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ between the CPU and APs via fronthaul links, the required fronthaul load is given by [14], [40], and [41]

$$\text{Fronthaul Load} = 2 \times m \times n \times N_Q \times f_s \text{ [bit/s]}, \tag{38}$$

where the factor of 2 accounts for the in-phase and quadrature components of the complex data, N_Q denotes the number of quantization bits per sample, and f_s is the sampling rate

TABLE II

INTERACTION COST AND COMPUTATIONAL COMPLEXITY

	Interaction Cost	**Computational Complexity**	
		Before Iteration	**Each Iteration**
WMMSE	$\sum_{i\in\mathcal{I}}\left(N_{\text{total}}+\sum_{k\in\mathcal{K}_i}D_{i,k}\right)M_i$	\	$\mathcal{O}\left({M_i}^3\right)$
RWMMSE	$\sum_{i\in\mathcal{I}}\left(\frac{1}{2}N_{\text{total}}^2+\sum_{k\in\mathcal{K}_i}D_{i,k}\min(M_i,N_{\text{total}})\right)$	$\mathcal{O}\left(M_iN_{\text{total}}^2\right)$	$\mathcal{O}(\min(M_i^3,N_{\text{total}}^2D_{\text{total}}))$
Local EZF	0	\	$\mathcal{O}\left(N_k^2M_i\right)$

measured in hertz. Similar to [41], N_Q and f_s are assumed fixed in this paper. Thus, we will evaluate the interaction cost by the size of the transmitted matrices, and the fronthaul load can be calculated via (38) accordingly.

In Fig. 2, we illustrate the interaction of the WMMSE and RWMMSE algorithms. Particularly, in the WMMSE algorithm, each AP i needs to transmit its raw channel matrix $\bar{\mathbf{H}}_i \in \mathbb{C}^{N_{\text{total}}\times M_i}$ to the CPU, and the CPU needs to transmit the beamformers $\{\mathbf{P}_{i,k}^\star \in \mathbb{C}^{M_i\times D_{i,k}} : k \in \mathcal{K}_i\}$ back to each AP. As a result, the interaction cost of the WMMSE algorithm is $\sum_{i\in\mathcal{I}}\left(N_{\text{total}}+\sum_{k\in\mathcal{K}_i}D_{i,k}\right)M_i$. In the RWMMSE algorithm, instead of the raw channel matrices $\{\bar{\mathbf{H}}_i\}$, the CPU only requires $\{\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H\}$, each has a size of $N_{\text{total}}\times N_{\text{total}}$. Furthermore, since $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$ is a symmetric matrix, AP i only needs to transmit the upper triangular part of $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$ to the CPU. The CPU calculates and transmits the low-dimensional matrices $\{\mathbf{X}_{i,k}^\star \in \mathbb{C}^{N_{\text{total}}\times D_{i,k}} : k \in \mathcal{K}_i\}$ when $M_i \geq N_{\text{total}}$, or $\{\breve{\mathbf{X}}_{i,k}^\star \in \mathbb{C}^{M_i\times D_{i,k}} : k \in \mathcal{K}_i\}$ when $M_i < N_{\text{total}}$, back to AP i. As such, the interaction cost of the RWMMSE algorithm is $\sum_{i\in\mathcal{I}}\left(\frac{1}{2}N_{\text{total}}^2+\sum_{k\in\mathcal{K}_i}D_{i,k}\min(M_i,N_{\text{total}})\right)$. It can be observed that as long as $M_i > \frac{1}{2}N_{\text{total}}$, the interaction cost of the RWMMSE algorithm is lower than that of the WMMSE algorithm. Finally, there is no interaction required for the local EZF, since it only uses local CSI for beamforming.

2) Computational Complexity: For the WMMSE algorithm, the dominate operation is computing the matrix inverse of $\left(\sum_{l\in\mathcal{K}}\alpha_l\mathbf{H}_{i,l}^H\mathbf{A}_l\mathbf{H}_{i,l}+\mu_i\mathbf{I}\right) \in \mathbb{C}^{M_i\times M_i}$ in the update of $\mathbf{P}_{i,k}^\star$ via (17), and the complexity is given by $\mathcal{O}(M_i^3)$. In the RWMMSE algorithm, updating $\mathbf{P}_{i,k}^\star$ is replaced by updating its low-dimensional substitution $\mathbf{X}_{i,k}^\star$ when $M_i \geq N_{\text{total}}$, or $\breve{\mathbf{X}}_{i,k}^\star$ when $M_i < N_{\text{total}}$. Specifically, the update of $\mathbf{X}_{i,k}^\star$ using (24) has a complexity of $\mathcal{O}(N_{\text{total}}^2D_{\text{total}})$, while updating $\breve{\mathbf{X}}_{i,k}^\star$ via (37) incurs a complexity of $\mathcal{O}(\min(M_i^3,N_{\text{total}}^2D_{\text{total}}))$. Hence, the complexity of the RWMMSE iteration process does not exceed that of the classic WMMSE iteration process. Moreover, before iteration process, the RWMMSE algorithm requires AP i to compute $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$, which yields a complexity of $\mathcal{O}(M_iN_{\text{total}}^2)$. In addition, when $M_i < N_{\text{total}}$, the calculation of the substitution $\breve{\mathbf{X}}_{i,k}^\star$ requires a reduced SVD of $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$ at the CPU, which also has a complexity of $\mathcal{O}(M_iN_{\text{total}}^2)$ [42]. Note that when $M_i < N_{\text{total}}$, although the complexity $\mathcal{O}(M_iN_{\text{total}}^2)$ in the RWMMSE is greater than the complexity $\mathcal{O}(M_i^3)$ in the WMMSE, $\bar{\mathbf{H}}_i\bar{\mathbf{H}}_i^H$ and its reduced SVD need to be calculated only once. Hence, the computational complexity of RWMMSE is comparable to that of WMMSE when $M_i < N_{\text{total}}$, unless $M_i \ll N_{\text{total}}$, which is uncommon in modern network configurations. And when $M_i > N_{\text{total}}$, RWMMSE demonstrates significantly greater computational efficiency. Finally, the dominant operation in the local EZF method is performing SVD on $\mathbf{H}_{i,k} \in \mathbb{C}^{N_k\times M_i}$ with a complexity of $\mathcal{O}(N_k^2M_i)$.

In conclusion, the local EZF method achieves the lowest computational complexity without any interaction. The computational complexity of the RWMMSE algorithm is *linear* in M_i, while that of the WMMSE algorithm is *cubic* in M_i. Additionally, the RWMMSE algorithm has lower interaction cost than the traditional WMMSE when $M_i > \frac{1}{2}N_{\text{total}}$, and significantly reduced computational complexity when $M_i > N_{\text{total}}$, by effectively utilizing the two low-dimensional substitutions. In the extreme case where $M_i < \frac{1}{2}N_{\text{total}}$, the standard WMMSE algorithm can be directly applied.

IV. JOINT BEAMFORMING AND STREAM ALLOCATION

In this section, we further explore the joint beamforming and DSA problem (3). Despite the efficient RWMMSE algorithm proposed in Section III, problem (3) is still NP-hard due to its mixed-integer and coupled decision variables. By effectively decoupling the decision variables $\{\mathbf{P}_{i,k}\}$ and $\{D_{i,k}\}$, and subtly relaxing the integer constraints, we propose a joint beamforming and linear-complexity stream allocation (LSA) method, termed as RWMMSE-LSA, to address problem (3).

A. Decoupling Beamforming and DSA

For ease of illustration, we rewrite problem (3) in a concise form as follows:

$$\max_{\{\mathbf{P}_{i,k}\},\{D_{i,k}\}} \sum_{k\in\mathcal{K}}\alpha_k\log\det\left(\mathbf{I}+\mathbf{H}_k\mathbf{P}_k\mathbf{P}_k^H\mathbf{H}_k^H\mathbf{N}_k^{-1}\right) \quad (39a)$$

$$\text{s.t.} \quad \sum_{k\in\mathcal{K}_i}\mathrm{Tr}\left(\mathbf{P}_{i,k}\mathbf{P}_{i,k}^H\right) \leq P_{\max,i}, \forall i, \quad (39b)$$

$$\sum_{i\in\mathcal{I}_k} D_{i,k} \leq N_k, \forall k, \quad (39c)$$

$$D_{i,k} \in \mathbb{N}, \forall k, \forall i. \quad (39d)$$

Recalling that the beamforming matrices $\{\mathbf{P}_{i,k}\}$ are highly coupled with the DSA parameters $\{D_{i,k}\}$, since the dimension of each $\mathbf{P}_{i,k}$ is determined by $D_{i,k}$. In order to decouple $\{\mathbf{P}_{i,k}\}$ and $\{D_{i,k}\}$, we introduce a *binary diagonal* stream indicator matrix $\mathbf{L}_{i,k} \in \mathbb{B}^{N_k\times N_k}$ for stream allocation, where zero diagonal entries indicate that the corresponding stream symbols are nulled. Furthermore, we introduce a *virtual* beamformer $\bar{\mathbf{P}}_{i,k} \in \mathbb{C}^{M_i\times N_k}$ which is expected to have some zero columns. With such notations, the actual beamformer $\mathbf{P}_{i,k}$ can be expressed as the matrix composed of all the non-zero columns of $\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}$, and the actual number of streams sent from AP i to UE k is $\mathrm{Tr}\left(\mathbf{L}_{i,k}\right) = D_{i,k}$.

Now we can recast problem (39) w.r.t. $\{\mathbf{P}_{i,k}\}$ and $\{D_{i,k}\}$ as

$$\max_{\substack{\{\bar{\mathbf{P}}_{i,k}\},\\ \{\mathbf{L}_{i,k}\}}} \sum_{k\in\mathcal{K}}\alpha_k\log\det\left(\mathbf{I}+\mathbf{H}_k\bar{\mathbf{P}}_k\mathbf{L}_k\mathbf{L}_k^H\bar{\mathbf{P}}_k^H\mathbf{H}_k^H\check{\mathbf{N}}_k^{-1}\right) \quad (40a)$$

$$\text{s.t.} \quad \sum_{k\in\mathcal{K}_i}\mathrm{Tr}\left(\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}\mathbf{L}_{i,k}^H\bar{\mathbf{P}}_{i,k}^H\right) \leq P_{\max,i}, \ \forall i, \quad (40b)$$

$$\sum_{i\in\mathcal{I}_k}\mathrm{Tr}(\mathbf{L}_{i,k}) \leq N_k, \ \forall k, \quad (40c)$$

$$\mathbf{L}_{i,k}(m) \in \{0,1\},\ \forall k,\ \forall i,\ m=1,\ldots,N_k, \qquad \text{(40d)}$$

where $\check{\mathbf{N}}_k \triangleq \sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\bar{\mathbf{P}}_{j,l}\mathbf{L}_{j,l}\mathbf{L}_{j,l}^H\bar{\mathbf{P}}_{j,l}^H\mathbf{H}_{j,k}^H+\sigma_k^2\mathbf{I}$, $\bar{\mathbf{P}}_k \triangleq \text{blkdiag}(\bar{\mathbf{P}}_{i_1,k},\ldots,\bar{\mathbf{P}}_{i_{|\mathcal{I}_k|},k}) \in \mathbb{C}^{M_{(k)}\times|\mathcal{I}_k|N_k}$, $\mathbf{L}_k \triangleq \text{blkdiag}(\mathbf{L}_{i_1,k},\ldots,\mathbf{L}_{i_{|\mathcal{I}_k|},k}) \in \mathbb{B}^{|\mathcal{I}_k|N_k\times|\mathcal{I}_k|N_k}$, and $\mathbf{L}_{i,k}(m)$ denotes the m-th diagonal element of $\mathbf{L}_{i,k}$.

By contradiction, it is easy to verify that the individual power constraints (40b) are equivalent to

$$\sum_{k\in\mathcal{K}_i}\text{Tr}(\bar{\mathbf{P}}_{i,k}\bar{\mathbf{P}}_{i,k}^H)\leq P_{\max,i}, \forall i. \qquad (41)$$

Indeed, since the actual beamformer consists of the non-zero columns of $\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}$, it is a waste to allocate transmit power to the columns of $\bar{\mathbf{P}}_{i,k}$, whose position is aligned with that of the zero columns of $\mathbf{L}_{i,k}$. Consequently, the optimal solution is pouring the transmit power into the columns of $\bar{\mathbf{P}}_{i,k}$ that correspond to the non-zero columns in $\mathbf{L}_{i,k}$.

As such, the original problem (39) with coupled variables $\{\mathbf{P}_{i,k}\}$ and $\{D_{i,k}\}$ is equivalent to the following problem with *independent* variables $\{\bar{\mathbf{P}}_{i,k}\}$ and $\{\mathbf{L}_{i,k}\}$:

$$\begin{aligned}\max_{\substack{\{\bar{\mathbf{P}}_{i,k}\},\\ \{\mathbf{L}_{i,k}\}}}\ & \sum_{k\in\mathcal{K}}\alpha_k\log\det\left(\mathbf{I}+\mathbf{H}_k\bar{\mathbf{P}}_k\mathbf{L}_k\mathbf{L}_k^H\bar{\mathbf{P}}_k^H\mathbf{H}_k^H\check{\mathbf{N}}_k^{-1}\right)\\ \text{s.t.}\ \ & \text{(40c)},\ \text{(40d)},\ \text{(41)}. \end{aligned}\qquad (42)$$

B. WMMSE Transformation and Problem Reformulation

Following the WMMSE framework, the WSR maximization problem (42) can be equivalently transformed to a weighted MSE minimization problem, i.e.,

$$\begin{aligned}\max_{\substack{\{\mathbf{U}_k\},\{\mathbf{W}_k\succ\mathbf{0}\},\\ \{\bar{\mathbf{P}}_{i,k}\},\{\mathbf{L}_{i,k}\}}}\ & \sum_{k\in\mathcal{K}}\alpha_k\left(\log\det(\mathbf{W}_k)-\text{Tr}\left(\mathbf{W}_k\mathbf{E}_k(\mathbf{U}_k,\bar{\mathcal{P}},\mathcal{L})\right)\right)\\ \text{s.t.}\ \ & \text{(40c)},\ \text{(40d)},\ \text{(41)},\end{aligned}\qquad (43)$$

where $\bar{\mathcal{P}} \triangleq \{\bar{\mathbf{P}}_{i,k}\}$, $\mathcal{L} \triangleq \{\mathbf{L}_{i,k}\}$, and $\mathbf{E}_k(\mathbf{U}_k,\bar{\mathcal{P}},\mathcal{L}) \triangleq (\mathbf{I}-\mathbf{U}_k^H\mathbf{H}_k\bar{\mathbf{P}}_k\mathbf{L}_k)(\mathbf{I}-\mathbf{U}_k^H\mathbf{H}_k\bar{\mathbf{P}}_k\mathbf{L}_k)^H+\mathbf{U}_k^H\check{\mathbf{N}}_k\mathbf{U}_k$. Here, the sizes of $\mathbf{U}_k$ and $\mathbf{W}_k$ are now given by $N_k\times|\mathcal{I}_k|N_k$ and $|\mathcal{I}_k|N_k\times|\mathcal{I}_k|N_k$, respectively.

A straightforward approach for solving (43) is to relax the integer constraints (40d) to box constraints within $[0,\ 1]$, thereby transforming (43) into a convex problem w.r.t. each of $\{\mathbf{U}_k\}$, $\{\mathbf{W}_k\}$, $\{\bar{\mathbf{P}}_{i,k}\}$, and $\{\mathbf{L}_{i,k}\}$, respectively, which can then be solved via the BCD method. However, this way cannot guarantee that the resulting solution satisfies the integer constraints (40d).

To address the challenges associated with integer constraints (40d), we introduce the following proposition, which indicates that the quadratic and mixed-integer problem (43) is equivalent to a continuous problem that is linear w.r.t $\{\mathbf{L}_{i,k}\}$. Based on this, the optimization problem becomes tractable.

Proposition 3: Problem (43) is equivalent to

$$\begin{aligned}\max_{\substack{\{\mathbf{U}_k\},\{\mathbf{W}_k\succ\mathbf{0}\},\\ \{\bar{\mathbf{P}}_{i,k}\},\{\mathbf{L}_{i,k}\}}}\ & \sum_{k\in\mathcal{K}}\alpha_k\left(\log\det(\mathbf{W}_k)+2\text{Tr}(\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_k\bar{\mathbf{P}}_k\mathbf{L}_k)\right)\\ & -\sum_{k\in\mathcal{K}}\alpha_k\text{Tr}\left(\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_k\bar{\mathbf{P}}_k\mathbf{L}_k\bar{\mathbf{P}}_k^H\mathbf{H}_k^H\mathbf{U}_k\right)\\ & -\sum_{k\in\mathcal{K}}\alpha_k\text{Tr}\left(\mathbf{W}_k\mathbf{U}_k^H\bar{\mathbf{N}}_k\mathbf{U}_k\right) \end{aligned}\qquad \text{(44a)}$$

$$\begin{aligned}\text{s.t.}\ & \text{(40c)}, \text{(41)},\\ & \mathbf{L}_{i,k}(m)\in[0,1], \forall k, \forall i, m=1,\ldots,N_k,\end{aligned}\qquad \text{(44b)}$$

where $\bar{\mathbf{N}}_k \triangleq \sum_{l\in\mathcal{K}_{-k}}\sum_{j\in\mathcal{I}_l}\mathbf{H}_{j,k}\bar{\mathbf{P}}_{j,l}\mathbf{L}_{j,l}\bar{\mathbf{P}}_{j,l}^H\mathbf{H}_{j,k}^H+\sigma_k^2\mathbf{I}$.

Proof: Since each $\mathbf{L}_{i,k}$ is a binary diagonal matrix, we have $\mathbf{L}_{i,k}\mathbf{L}_{i,k}^H=\mathbf{L}_{i,k}, \forall k, \forall i$. This *quadratic-linear* identity implies that the objective of (43) is equivalent to the objective (44a), which is linear w.r.t $\{\mathbf{L}_{i,k}\}$. Moreover, since the constraints of $\{\mathbf{L}_{i,k}\}$ are independent of the other variables, the optimal $\{\mathbf{L}_{i,k}^\star\}$ of problem (44) are located at the vertices of the feasible region, which are either 0 or 1. Consequently, the 0-1 integer constraints (40d) can be equivalently transformed to the continuous constraints (44b). ■

C. RWMMSE-LSA Algorithm

Toward this end, problem (44) is convex w.r.t. each of the decision variables $\{\mathbf{W}_k\}$, $\{\mathbf{U}_k\}$, $\{\bar{\mathbf{P}}_{i,k}\}$ and $\{\mathbf{L}_{i,k}\}$. In the following, we use the BCD method together with the low-dimensional subspace property to provide an efficient joint beamforming and DSA algorithm.

1) Optimizing $\{\mathbf{L}_{i,k}\}$ with fixed $\{\mathbf{W}_k\}$, $\{\mathbf{U}_k\}$ and $\{\bar{\mathbf{P}}_{i,k}\}$: the linear DSA problem w.r.t. $\{\mathbf{L}_{i,k}\}$ is

$$\begin{aligned}\min_{\{\mathbf{L}_{i,k}\}}\ & \sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{I}_k}\alpha_k\text{Tr}\left(\bar{\mathbf{P}}_{i,k}^H\mathbf{H}_{i,k}^H\mathbf{A}_k\mathbf{H}_{i,k}\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}\right)\\ & -2\sum_{k\in\mathcal{K}}\sum_{i\in\mathcal{I}_k}\alpha_k\text{Re}\{\text{Tr}\left(\bar{\boldsymbol{\Xi}}_{i,k}\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_{i,k}\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}\right)\}\\ & +\sum_{k\in\mathcal{K}}\alpha_k\text{Tr}\left(\sum_{j\in\mathcal{I}_l}\sum_{l\in\mathcal{K}_{-k}}(\bar{\mathbf{P}}_{j,l})^H\mathbf{H}_{j,k}^H\mathbf{A}_k\mathbf{H}_{j,k}\bar{\mathbf{P}}_{j,l}\mathbf{L}_{j,l}\right)\\ \text{s.t.}\ & \text{(40c)},\ \text{(44b)},\end{aligned}\qquad (45)$$

where $\bar{\boldsymbol{\Xi}}_{i,k}$ is a $N_k\times|\mathcal{I}_k|N_k$ binary matrix with elements from the $((i-1)\times N_k+1)$-th column to the $(i\times N_k)$-th column being $\mathbf{I}_{N_k}$ and the rest elements being zero.

Note that the update of $\{\mathbf{L}_{i,k}\}$ can be decoupled across UEs, resulting in the following optimization problem:

$$\begin{aligned}\min_{\{\mathbf{L}_{i,k},i\in\mathcal{I}_k\}}\ & \sum_{i\in\mathcal{I}_k}\sum_{l\in\mathcal{K}}\alpha_l\text{Tr}\left(\bar{\mathbf{P}}_{i,k}^H\mathbf{H}_{i,l}^H\mathbf{A}_l\mathbf{H}_{i,l}\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}\right)\\ & -2\sum_{i\in\mathcal{I}_k}\alpha_k\text{Re}\left\{\text{Tr}\left(\bar{\boldsymbol{\Xi}}_{i,k}\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_{i,k}\bar{\mathbf{P}}_{i,k}\mathbf{L}_{i,k}\right)\right\}\\ \text{s.t.}\ & \text{(40c)},\ \text{(44b)},\end{aligned}\qquad (46)$$

for each k. More concisely, problem (46) can be expressed as a linear programming problem, i.e.,

$$\begin{aligned}\min_{\mathbf{L}_k}\ & \text{Tr}(\boldsymbol{\Psi}_k\mathbf{L}_k)\\ \text{s.t.}\ & \mathbf{L}_k(l)\in[0,1],\ \forall l=1,\ldots,|\mathcal{I}_k|N_k,\\ & \text{Tr}(\mathbf{L}_k)\leq N_k,\end{aligned}\qquad (47)$$

where $\mathbf{L}_k(l)$ denotes the l-th diagonal element of $\mathbf{L}_k$, $\boldsymbol{\Psi}_k \triangleq \text{blkdiag}(\text{diag}(\boldsymbol{\Psi}_{i_1,k}),\ldots,\text{diag}(\boldsymbol{\Psi}_{i_{|\mathcal{I}_k|},k}))$ with $\boldsymbol{\Psi}_{i,k}$ being

$$\begin{aligned}\boldsymbol{\Psi}_{i,k} \triangleq & \sum_{l\in\mathcal{K}}\alpha_l\bar{\mathbf{P}}_{i,k}^H\mathbf{H}_{i,l}^H\mathbf{A}_l\mathbf{H}_{i,l}\bar{\mathbf{P}}_{i,k}\\ & -2\alpha_k\text{Re}\left\{\bar{\boldsymbol{\Xi}}_{i,k}\mathbf{W}_k\mathbf{U}_k^H\mathbf{H}_{i,k}\bar{\mathbf{P}}_{i,k}\right\}.\end{aligned}\qquad (48)$$

It is evident that the optimal solution to problem (47) has a closed form, which is selecting the negative diagonal elements of $\boldsymbol{\Psi}_k$. If there are more than N_k negative elements, the N_k smallest ones are chosen. To be more specific, the optimal solution $\mathbf{L}_k^\star$ to problem (47) is

$$\mathbf{L}_k^\star(l) = \begin{cases} 1, & \text{if } \boldsymbol{\Psi}_k(l) < 0 \text{ and } |\{j : \boldsymbol{\Psi}_k(j) < \boldsymbol{\Psi}_k(l)\}| \le N_k - 1, \\ 0, & \text{otherwise,} \end{cases} \tag{49}$$

where $\boldsymbol{\Psi}_k(l)$ denotes the l-th diagonal element of $\boldsymbol{\Psi}_k$.

2) Optimizing $\{\bar{\mathbf{P}}_{i,k}\}$ with fixed $\{\mathbf{W}_k\}$, $\{\mathbf{U}_k\}$ and $\{\mathbf{L}_{i,k}\}$: by dropping the constant terms, problem (44) is reduced to

$$\begin{aligned} \min_{\{\bar{\mathbf{P}}_{i,k}\}} & \sum_{k\in\mathcal{K}} \alpha_k \sum_{i\in\mathcal{I}_k} \mathrm{Tr}\left(\bar{\mathbf{P}}_{i,k}^H \mathbf{H}_{i,k}^H \mathbf{A}_k \mathbf{H}_{i,k} \bar{\mathbf{P}}_{i,k} \mathbf{L}_{i,k}\right) \\ & - 2\sum_{k\in\mathcal{K}} \alpha_k \sum_{i\in\mathcal{I}_k} \mathrm{Re}\left\{\mathrm{Tr}\left(\bar{\boldsymbol{\Xi}}_{i,k} \mathbf{W}_k \mathbf{U}_k^H \mathbf{H}_{i,k} \bar{\mathbf{P}}_{i,k} \mathbf{L}_{i,k}\right)\right\} \\ & + \sum_{k\in\mathcal{K}} \alpha_k \mathrm{Tr}\left(\mathbf{A}_k \sum_{l\in\mathcal{K}_{-k}} \sum_{j\in\mathcal{I}_l} \mathbf{H}_{j,k} \bar{\mathbf{P}}_{j,l} \mathbf{L}_{j,l} \bar{\mathbf{P}}_{j,l}^H \mathbf{H}_{j,k}^H\right) \\ \text{s.t. } & (41). \end{aligned} \tag{50}$$

Similar to Section III-B, $\{\bar{\mathbf{P}}_{i,k}\}$ can be updated either by directly solving problem (50) or by using the low-dimensional subspace property in Theorem 1 when $M_i \ge N_{\text{total}}$, or in Corollary 2 when $M_i < N_{\text{total}}$. The former yields

$$\bar{\mathbf{P}}_{i,k}^\star = \left(\sum_{l\in\mathcal{K}} \alpha_l \mathbf{H}_{i,l}^H \mathbf{A}_l \mathbf{H}_{i,l} + \mu_i \mathbf{I}\right)^{-1} \alpha_k \mathbf{H}_{i,k}^H \mathbf{U}_k \mathbf{W}_k \bar{\boldsymbol{\Xi}}_{i,k}^H \mathbf{L}_{i,k}, \tag{51}$$

while the latter updates $\bar{\mathbf{P}}_{i,k}$, when $M_i \ge N_{\text{total}}$, via the equality

$$\bar{\mathbf{P}}_{i,k}^\star \mathbf{L}_{i,k} = \bar{\mathbf{H}}_i^H \bar{\mathbf{X}}_{i,k}^\star, \tag{52}$$

where $\bar{\mathbf{X}}_{i,k}^\star \in \mathbb{C}^{N_{\text{total}} \times N_k}$ is a low-dimensional substitution for $\bar{\mathbf{P}}_{i,k}^\star \mathbf{L}_{i,k}$, and is given by

$$\bar{\mathbf{X}}_{i,k}^\star = \mathbf{U}\mathbf{W}\left(\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H \mathbf{U}\mathbf{W} + \mu_i \mathbf{I}\right)^{-1} \bar{\mathbf{K}}_{i,k} \bar{\boldsymbol{\Xi}}_{i,k}^H \mathbf{L}_{i,k}, \tag{53}$$

where $\bar{\mathbf{K}}_{i,k} \triangleq \left[\mathbf{0}_{N_{(k)} \times \sum_{l=1}^{k-1} N_{(l)}}, \ \mathbf{I}_{N_{(k)}}, \ \mathbf{0}_{N_{(k)} \times \sum_{l=k+1}^{K} N_{(l)}}\right]^H$ with $N_{(k)} \triangleq |\mathcal{I}_k| N_k$. The derivation of the corresponding substitution for $\bar{\mathbf{P}}_{i,k}^\star \mathbf{L}_{i,k}$ when $M_i < N_{\text{total}}$ is similar to that of $\bar{\mathbf{X}}_{i,k}^\star$, and are thus omitted. Hereafter, we still choose the common condition $M_i \ge N_{\text{total}}$ to illustrate the proposed joint beamforming and DSA algorithm. In the latter update (52), each column of $\bar{\mathbf{P}}_{i,k}^\star$ is set to zero if its corresponding diagonal element in $\mathbf{L}_{i,k}$ is 0. Conversely, if a diagonal element in $\mathbf{L}_{i,k}$ is 1, the corresponding column in $\bar{\mathbf{P}}_{i,k}^\star$ is given by the corresponding column of $\bar{\mathbf{H}}_i^H \bar{\mathbf{X}}_{i,k}^\star$. Similar to the discussion in Section III-C, updating $\bar{\mathbf{P}}_{i,k}^\star$ via (52) has a lower computational complexity, since the dimension of $\bar{\mathbf{X}}_{i,k}^\star$ is smaller than that of $\bar{\mathbf{P}}_{i,k}^\star$. Finally, the actual beamformer $\mathbf{P}_{i,k}$ is obtained by selecting the non-zero columns of $\bar{\mathbf{P}}_{i,k}^\star \mathbf{L}_{i,k}^\star$.

3) Optimizing $\{\mathbf{U}_k\}$ and $\{\mathbf{W}_k\}$ with fixed $\{\bar{\mathbf{P}}_{i,k}\}$ and $\{\mathbf{L}_{i,k}\}$: by substituting (52) into (14) and (15), $\{\mathbf{U}_k\}$ and $\{\mathbf{W}_k\}$ are updated as, respectively,

$$\begin{aligned} \mathbf{U}_k^\star = & \left(\check{\mathbf{N}}_k + \mathbf{H}_k \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \bar{\mathbf{X}}_{\mathcal{I}_k} \bar{\mathbf{X}}_{\mathcal{I}_k}^H \tilde{\mathbf{H}}_{\mathcal{I}_k} \mathbf{H}_k^H\right)^{-1} \\ & \times \mathbf{H}_k \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \bar{\mathbf{X}}_{\mathcal{I}_k}, \ \forall k, \end{aligned} \tag{54}$$

$$\mathbf{W}_k^\star = \left(\mathbf{I} - (\mathbf{U}_k^\star)^H \mathbf{H}_k \tilde{\mathbf{H}}_{\mathcal{I}_k}^H \bar{\mathbf{X}}_{\mathcal{I}_k}\right)^{-1}, \ \forall k, \tag{55}$$

where $\bar{\mathbf{X}}_{\mathcal{I}_k} \triangleq \text{blkdiag}(\bar{\mathbf{X}}_{i_1,k}, \ldots, \bar{\mathbf{X}}_{i_{|\mathcal{I}_k|},k})$.

Algorithm 4 Joint Beamforming and DSA Algorithm (RWMMSE-LSA)

1: Each AP i calculates $\bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H$;
   Interaction: Each AP i sends $\bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H$ to the CPU pool;
2: Initialize $\{\mathbf{L}_{i,k}\}$ and $\{\bar{\mathbf{X}}_{i,k}\}$;
3: **Repeat**
4: Update $\{\mathbf{U}_k^\star\}$ via (54);
5: Update $\{\mathbf{W}_k^\star\}$ via (55);
6: Update $\{\bar{\mathbf{X}}_{i,k}^\star\}$ via (53);
7: Update $\{\mathbf{L}_{i,k}^\star\}$ via (49);
8: **Until** Convergence
   Interaction: Send $\{\mathbf{X}_{i,k}^\star : k \in \mathcal{K}_i\}$ to each AP i;
Output: Each AP i calculates $\{\mathbf{P}_{i,k}^\star = \bar{\mathbf{H}}_i^H \mathbf{X}_{i,k}^\star : k \in \mathcal{K}_i\}$.

Algorithm 4 summarizes the proposed joint beamforming and DSA algorithm. Notably, compared to the pure RWMMSE beamforming strategy in Algorithm 2, the additional cost for Algorithm 4 is only the computation of (49), which entails significantly low computational complexity.

Remark 3: (AP-UE Pairing): The RWMMSE-LSA algorithm is adaptable for AP-UE pairing. Specifically, by initially setting $\mathcal{I}_k = \mathcal{I}$ for each UE k before executing Algorithm 4, the RWMMSE-LSA algorithm selects the channels with the best conditions from *all* APs. Consequently, each AP i corresponding to the selected channels is designated as a serving AP for UE k, i.e., $\mathcal{I}_k = \{i : D_{i,k} > 0\}$. This modified approach is referred to as RWMMSE-LUS.

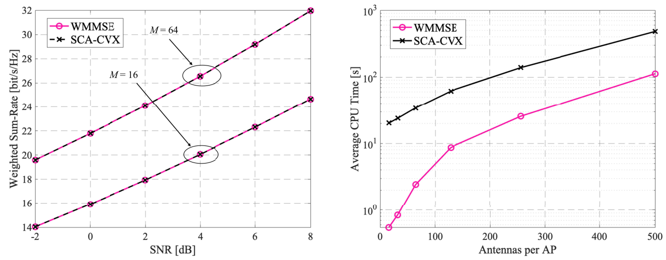
The convergence of the RWMMSE-LSA algorithm is established in Proposition 4.

Proposition 4: Any limit point $(\mathbf{U}_k^\star, \mathbf{W}_k^\star, \bar{\mathbf{P}}_{i,k}^\star, \mathbf{L}_{i,k}^\star)$ of the iterates generated by the RWMMSE-LSA algorithm is a stationary point of the relaxed problem (44), and the corresponding $(\mathbf{P}_{i,k}^\star, D_{i,k}^\star)$ is a feasible solution of its original equivalent problem (3) with $\mathbf{P}_{i,k}^\star$ comprising the non-zero columns of $\bar{\mathbf{P}}_{i,k}^\star \mathbf{L}_{i,k}^\star$ and $D_{i,k}^\star = \mathrm{Tr}(\mathbf{L}_{i,k}^\star)$.

Proof: See Appendix D. ■

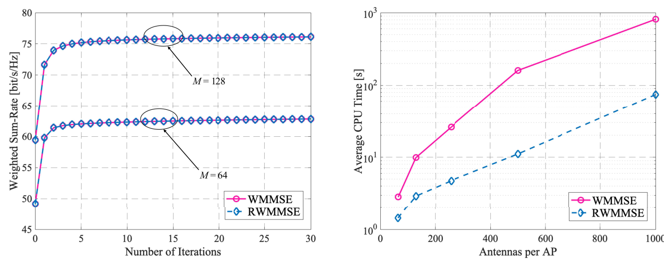
V. Simulation Results

In this section, we present simulation results to illustrate the efficiency of the proposed algorithms. We model the channel as a Rayleigh fading channel with CSCG distribution. The pathloss is modeled as $\beta_{i,k} = 128.1 + 37.6\log_{10}(d_{i,k})$[dB] [43], where $d_{i,k} \in [0, 0.3]$ in kilometers denotes the distance between AP i and UE k. Without loss of generality, all APs (and UEs) are assumed to have the same number of antennas, i.e., $M_i = M$ and $N_k = N$. The transmit power budget is set to be $P_{\max} = 10^{\frac{\text{SNR}}{10}}$ for all APs. For each UE k, its fairness



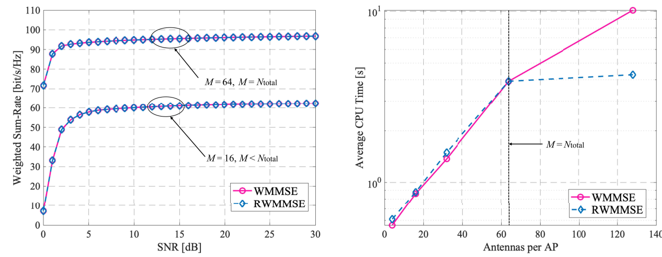
(a) (b)

Fig. 3. Comparison of the SCA-CVX algorithm in [4] and the WMMSE algorithm: (a) WSR, (b) average CPU time when SNR = 6 dB.



(a) (b)

Fig. 4. Comparison of the WMMSE and RWMMSE algorithms for $M_i \geq N_{\text{total}}$: (a) WSR, (b) average CPU time when SNR = 6 dB.



(a) (b)

Fig. 5. Comparison of the WMMSE and RWMMSE algorithms for $M_i \leq N_{\text{total}}$: (a) WSR, (b) average CPU time when SNR = 6 dB.

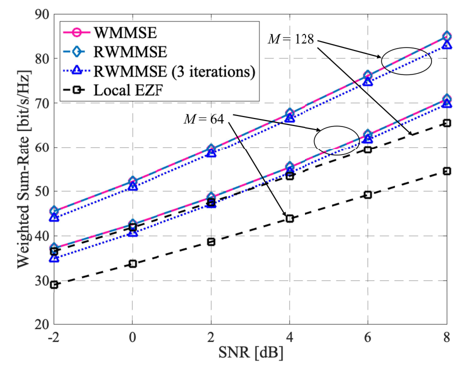


Fig. 6. Comparison of WSR among the fully distributed local EZF method, the WMMSE algorithm and the RWMMSE algorithm.

weight is set as $\alpha_k = 1$, its receiving noise power density is -100 dBm/Hz with operation bandwidth being 1 MHz. In addition, when evaluating the pure beamforming algorithms, we assume that each serving AP transmits the same number of data streams to UE k, i.e., $D_{i,k} = D$, and each UE selects the L nearest APs as its serving set, i.e., $|\mathcal{I}_k| = L$. Unless otherwise stated, the system parameters (I, K, L, M, N, D) are set as $(4, 8, 2, 64, 4, 2)$. Monte Carlo simulations are performed over 100 randomly generated channel realizations. The local EZF beamformer is used to initialize the proposed algorithms.

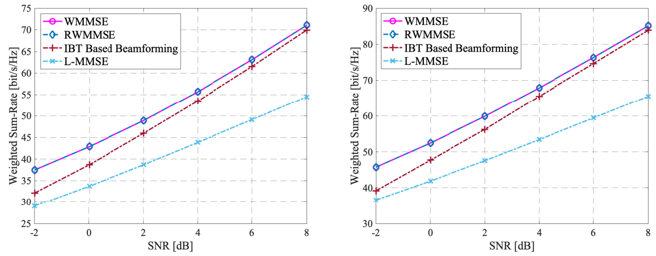
We evaluate the performance of the proposed algorithms for the pure beamforming problem (4) and the joint beamforming and DSA problem (3) in subsections V-A and V-B, respectively. Subsection V-C extends the simulations to the imperfect CSI model and the spatially correlated channel model. Finally, subsection V-D compares the performance of the proposed algorithms in NCJT and CJT scenarios.

A. *Performance Evaluation for the Pure Beamforming Problem (4)*

In Fig. 3, we compare the WMMSE algorithm and the SCA-CVX algorithm proposed in [4]. We set $N = 1$ and $D = 1$, as the SCA-CVX method was proposed for the scenario where each UE has single antenna. Fig. 3 reveals that the SCA-CVX algorithm and the WMMSE algorithm achieve the same WSR, but the SCA-CVX algorithm consumes 100x+ time than the WMMSE algorithm to converge. Consequently, we choose the WMMSE algorithm as the baseline in the simulations below. Fig. 4 compares the RWMMSE algorithm and the WMMSE algorithm. Fig. 4(a) shows that both algorithms converge smoothly to the same WSR, which is in accordance with Theorem 1. Moreover, Fig. 4(b) shows that the WMMSE algorithm generally requires more CPU time than the RWMMSE algorithm, particularly as the number of per-AP antennas increases. This is because the WMMSE and the RWMMSE algorithms have linear and cubic complexity in M, respectively. Fig. 4 demonstrates that the RWMMSE algorithm achieves the same WSR performance as the WMMSE algorithm, but entails significantly lower computational complexity.

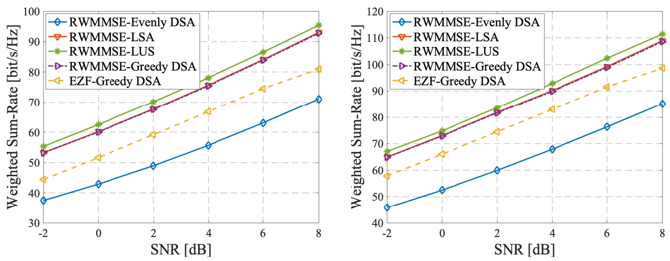
In Fig. 5, we examine the WSR and CPU time of the WMMSE and RWMMSE algorithms under the condition $M_i \leq N_{\text{total}}$. Specifically, we increase the number of UEs by setting $K = 16$, which corresponds to $N_{\text{total}} = 64$. Fig. 5(a) illustrates that RWMMSE achieves the same WSR as the traditional WMMSE for $M_i \leq N_{\text{total}}$. Furthermore, Fig. 5(b) indicates that when $M_i \leq N_{\text{total}}$, RWMMSE consumes a comparable CPU time as WMMSE. However, as the number of AP antennas M_i increases to $M_i > N_{\text{total}}$, RWMMSE requires significantly less time than WMMSE to converge. Consequently, Fig. 4 and Fig. 5 demonstrate the applicability of the proposed RWMMSE algorithm for any positive values of M_i, and exhibit significantly improved computational efficiency, especially in massive MIMO networks where $M_i \gg N_{\text{total}}$.

In Fig. 6, we compare the WSR of the proposed beamforming algorithms for different SNR, when $M = 64$ or 128. As expected, the WMMSE and the RWMMSE algorithms yield the same WSR, both significantly outperforming the local EZF method. Additionally, our RWMMSE algorithm achieves comparable WSR performance even within 3 iter-



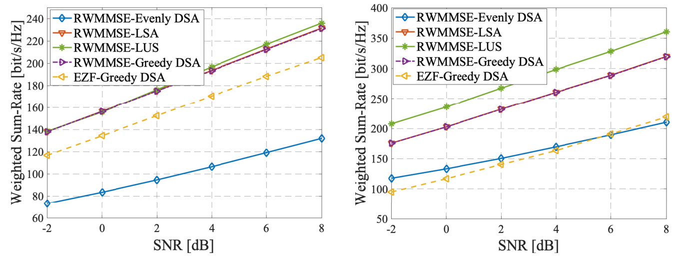
(a) $M = 64$ (b) $M = 128$

Fig. 7. WSR performance of different distributed beamforming algorithms.



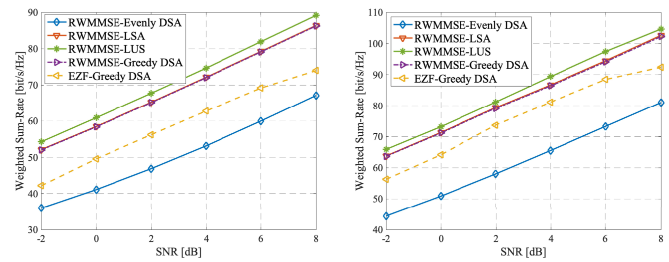
(a) $M = 64$ (b) $M = 128$

Fig. 8. WSR performance of different algorithms with $(I, K, L, N) = (4, 8, 2, 4)$.



(a) $K = 8$ (b) $K = 16$

Fig. 9. WSR performance of different algorithms with $(I, M, L, N) = (8, 64, 4, 8)$.



(a) $M = 64$ (b) $M = 128$

Fig. 10. WSR performance of different algorithms under CE errors.

ations, demonstrating its strong potential for deployment in computationally limited scenarios.

Fig. 7 presents the WSR of the WMMSE method, the RWMMSE method, the IBT based algorithm [21], [22], and the L-MMSE method [19], [20]. As observed, both the IBT-based and the L-MMSE beamforming methods suffer WSR loss due to partial CSI acquisition, while our RWMMSE demonstrates identical WSR to that of the classic WMMSE.

B. Performance Evaluation for the Joint Beamforming and DSA Problem (3)

In this subsection, we present simulation results for the joint beamforming and DSA optimization problem (3).

In Fig. 8, we compare the WSR of the pure beamforming RWMMSE algorithm with evenly DSA (named RWMMSE-Evenly DSA), the RWMMSE-LSA algorithm, the RWMMSE-LUS algorithm, and the EZF method integrated with greedy-based DSA proposed in [29] (named EZF-Greedy DSA), under different values of SNR. Additionally, we also incorporate our RWMMSE beamforming with the greedy-based DSA method [29] (named RWMMSE-Greedy DSA), and present its WSR performance. Here, we employ the DSA method in [29] based on the criterion of increasing the WSR most. As expected, the RWMMSE-LSA algorithm achieves higher WSR than the RWMMSE-Evenly DSA algorithm. Moreover, the RWMMSE-Greedy DSA algorithm achieves almost the same WSR as the RWMMSE-LSA, but the former requires an exhaustive search in each additional DSA step, which is of high complexity. Additionally, even the EZF-Greedy DSA has higher WSR than the RWMMSE-Evenly DSA, showing the significance of DSA. Furthermore, since the RWMMSE-LUS algorithm selects the most beneficial data streams among all the APs, it achieves the highest WSR.

In Fig. 9, we compare the performance of these algorithms in a larger-scale network with $(I, M, L, N) = (8, 64, 4, 8)$. As the network scale expands, whether by increasing the number of UEs (from Fig. 9(a) to Fig. 9(b)) or the number of (I, L, N) (from Fig. 8(a) to Fig. 9(a)), the performance gain of our RWMMSE-LSA and RWMMSE-LUS algorithms significantly outperforms that of the EZF-Greedy DSA algorithm. Moreover, the RWMMSE-LUS, as expected, achieves higher WSR than the RWMMSE-LSA, implying the importance of joint AP-UE pairing and DSA in MU-MIMO networks.

C. Performance Evaluation in Additional Channel Models

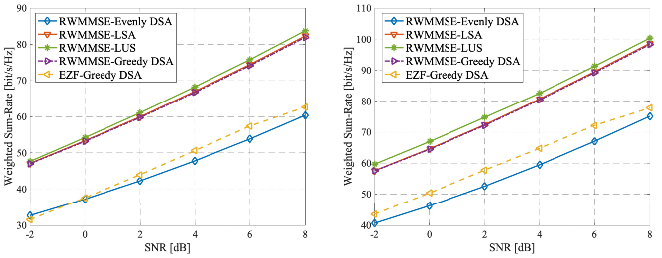
This subsection presents additional results under typical conditions of imperfect CSI and spatially correlated channels.

1) Imperfect CSI: One typical example of imperfect CSI arises from channel estimation (CE) errors. In this context, we use the CE case to evaluate the proposed algorithms, and the channel matrix can be expressed as [44], [45]

$$\mathbf{H}_{i,k} = \sqrt{1-\sigma_\epsilon^2}\hat{\mathbf{H}}_{i,k}^e + \tilde{\mathbf{H}}_{i,k}^e, \tag{56}$$

where σ_ϵ^2 is the variance of the channel estimation error, $\hat{\mathbf{H}}_{i,k}^e$ denotes the channel estimate, and $\tilde{\mathbf{H}}_{i,k}^e$ represents the channel estimation error. The elements of $\tilde{\mathbf{H}}_{i,k}^e$ are typically modeled as independent and identically distributed (i.i.d.) $\mathcal{CN}(0, \sigma_\epsilon^2)$.

In Fig. 10, we examine the impact of CE errors on the WSR produced by our proposed algorithms. We set $\sigma_\epsilon^2 = 0.01$, as CE errors typically remain below this threshold at medium SNRs during the channel estimation process [45]. Moreover, the beamformers $\{\mathbf{P}_{i,k}\}$ are computed based on the estimated channel matrices $\{\hat{\mathbf{H}}_{i,k}^e\}$, while the WSR is evaluated using the actual channel matrices $\{\mathbf{H}_{i,k}\}$. The WSR achieved by



(a) $M = 64$ (b) $M = 128$

Fig. 11. WSR performance of different algorithms under spatially correlated channels.

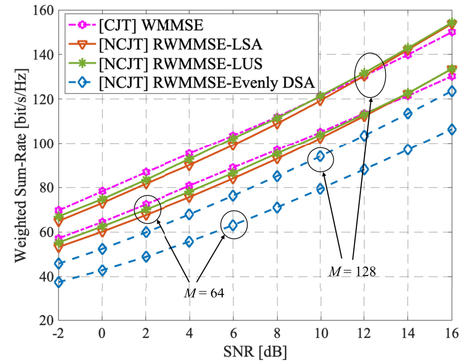


Fig. 12. Comparison of CJT and NCJT with different numbers of antennas per AP ($M = 64, 128$).

RWMMSE-Evenly DSA, RWMMSE-LSA, and RWMMSE-LUS in the presence of CE errors is close to that obtained with perfect CSI shown in Fig. 8. This highlights the effectiveness of the proposed methods under conditions of CE errors.

2) Spatially Correlated Channels: we adopt the jointly correlated model [46], also known as the Weichselberger model [47], which is given as follows:

$$\mathbf{H}_{i,k} = \mathbf{G}_{i,k,r}\left(\mathbf{M}_{i,k} \odot \mathbf{H}_{i,k,iid}\right)\mathbf{G}_{i,k,t}, \tag{57}$$

where $\odot$ denotes the element-wise product, $\mathbf{H}_{i,k,iid}$ is composed of i.i.d. $\mathcal{CN}(0,1)$ random entries, $\mathbf{G}_{i,k,r}$ and $\mathbf{G}_{i,k,t}$ denote the eigenvector matrices of the one-sided correlation matrices $\mathbf{R}_{i,k,r} \triangleq \mathbb{E}[\mathbf{H}_{i,k}\mathbf{H}_{i,k}^H]$ and $\mathbf{R}_{i,k,t} \triangleq \mathbb{E}[\mathbf{H}_{i,k}^T\mathbf{H}_{i,k}^*]$, respectively. Moreover, $\mathbf{M}_{i,k}$ is the element-wise square root of $\tilde{\mathbf{M}}_{i,k} \triangleq \mathbf{M}_{i,k} \odot \mathbf{M}_{i,k}$, which captures the joint correlation properties between AP i and UE k, illustrating the spatial arrangement of scattering objects.

In the simulations, we generate $\mathbf{R}_{i,k,r}$ and $\mathbf{R}_{i,k,t}$ using the exponential correlation model. Specifically, the (i,j)-th entries of $\mathbf{R}_{i,k,r}$ and $\mathbf{R}_{i,k,t}$ are given by $\rho_r^{|i-j|}$ and $\rho_t^{|i-j|}$, with $\rho_r = 0.3$ and $\rho_t = 0.8$, respectively. The coupling matrix $\tilde{\mathbf{M}}_{i,k}$ is given by [46]

$$\tilde{\mathbf{M}}_{i,k} = \beta_{i,k}\begin{bmatrix} \frac{M_i N_k}{2} & a\mathbf{1}_{1\times(M_i-1)} \\ a\mathbf{1}_{(N_k-1)\times 1} & a\mathbf{1}_{(N_k-1)\times(M_i-1)} \end{bmatrix} \tag{58}$$

with $a = \frac{M_i N_k}{2(M_i N_k - 1)}$.

Fig. 11 illustrates the WSR performance of the proposed algorithms in spatially correlated channels. As anticipated, the proposed algorithms remain applicable. The results, presented in Fig. 8 to Fig.11, demonstrate that our approaches are effective across various channel models. This is expected, as the derivation of the proposed algorithms does not assume any specific distribution for the channel matrices.

D. Comparison Between NCJT and CJT

Finally, Fig. 12 illustrates the performance gaps between CJT and NCJT scenarios under different numbers of transmit antennas. In CJT, serving APs $\mathcal{I}_k$ transmit the same data symbol to UE k. For a fair comparison, the number of transmit streams in CJT is set as $D = 4$. The WMMSE algorithm in CJT also utilizes the BCD method to alternately optimize the beamforming matrices $\{\mathbf{P}_{i,k,\text{CJT}}\}$ and the auxiliary variables $\{\mathbf{U}_{k,\text{CJT}}\}$ and $\{\mathbf{W}_{k,\text{CJT}}\}$. The updates are given by

$$\mathbf{U}_{k,\text{CJT}} = \left(\mathbf{N}_{k,\text{CJT}} + \mathbf{H}_k\mathbf{P}_{k,\text{CJT}}\mathbf{P}_{k,\text{CJT}}^H\mathbf{H}_k^H\right)^{-1}\mathbf{H}_k\mathbf{P}_{k,\text{CJT}}, \forall k, \tag{59a}$$

$$\mathbf{W}_{k,\text{CJT}} = \left(\mathbf{I} - \mathbf{U}_{k,\text{CJT}}^H\mathbf{H}_k\mathbf{P}_{k,\text{CJT}}\right)^{-1}, \forall k, \tag{59b}$$

$$\begin{aligned}\mathbf{P}_{i,k,\text{CJT}} = &\left(\sum_{l\in\mathcal{U}}\alpha_l\mathbf{H}_{i,l}^H\mathbf{A}_{l,\text{CJT}}\mathbf{H}_{i,l} + \mu_i\mathbf{I}\right)^{-1} \\ &\times\Bigg(\alpha_k\mathbf{H}_{i,k}^H\mathbf{U}_{k,\text{CJT}}\mathbf{W}_{k,\text{CJT}} \\ &\quad - \sum_{l\in\mathcal{U}}\sum_{j\neq i, j\in\mathcal{I}_k}\alpha_l\mathbf{H}_{i,l}^H\mathbf{A}_{l,\text{CJT}}\mathbf{H}_{j,l}\mathbf{P}_{j,k,\text{CJT}}\Bigg), \forall i, \forall k,\end{aligned} \tag{59c}$$

where $\mathbf{P}_{k,\text{CJT}} \triangleq [\mathbf{P}_{i_1,k}^H, \ldots, \mathbf{P}_{i_{|\mathcal{I}_k|},k}^H]^H$, and $\mathbf{A}_{l,\text{CJT}} \triangleq \mathbf{U}_{l,\text{CJT}}\mathbf{W}_{l,\text{CJT}}\mathbf{U}_{l,\text{CJT}}^H$.

As illustrated in Fig. 12, when the SNR increases, the WSR of NCJT that adopts the proposed joint beamforming and DSA methods, namely RWMMSE-DSA and RWMMSE-LUS gradually surpasses the WSR achieved by CJT using the WMMSE method. The simulation results indicate that CJT achieves higher data rates than NCJT with optimized DSA in low SNR regimes, owing to its superior coherent combining gain, which effectively enhances the received signal power in noise-dominated scenarios. However, as SNR increases, NCJT outperforms CJT due to the optimized stream allocation strategy that improves spatial multiplexing efficiency by concentrating power on high-quality channels while mitigating interference from weaker streams. In contrast, the fixed stream allocation in CJT inevitably wastes power on suboptimal channels, despite WMMSE-based power shaping.

VI. Conclusion

This work has investigated joint beamforming and DSA optimization for WSR maximization in NCJT. By fixing the number of transmit streams, we have first studied the low-dimensional substitution of the WMMSE beamformer in the pure beamforming problem. Based on this, a distributed RWMMSE beamforming algorithm has been proposed, which is equivalent to the original WMMSE algorithm but exhibits lower complexity and reduced interaction cost in massive MIMO systems. In addition, a local EZF method without any data interaction has been explored to initialize the RWMMSE iteration. Inspired by these algorithms, we have then proposed an RWMMSE-LSA algorithm that jointly optimizes beamforming and stream allocation. This algorithm converges to stationary points with closed-form updates. Simulations have demonstrated the significant advantages of our proposed algorithms over existing alternatives, in terms of both computational complexity and interaction cost.

Appendix A
Proof of Theorem 1

The update (19) is a stationary point of the beamforming optimization problem (4), and vice versa [16], [36]. By substituting $\mathbf{H}_{i,k}^H = \bar{\mathbf{H}}_i^H \mathbf{T}_{i,k}$ into the WMMSE update (19), $\mathbf{P}_{i,k}^\star$ can be rewritten as

$$\begin{aligned}\mathbf{P}_{i,k}^\star &= \left(\bar{\mathbf{H}}_i^H \mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i + \mu_i \mathbf{I}\right)^{-1} \alpha_k \bar{\mathbf{H}}_i^H \mathbf{T}_{i,k} \mathbf{U}_k^\star \mathbf{W}_k^\star \boldsymbol{\Xi}_{i,k}^H \\ &\overset{(a)}{=} \bar{\mathbf{H}}_i^H \underbrace{\alpha_k \left(\mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H + \mu_i \mathbf{I}\right)^{-1} \mathbf{T}_{i,k} \mathbf{U}_k^\star \mathbf{W}_k^\star \boldsymbol{\Xi}_{i,k}^H}_{\mathbf{X}_{i,k}^\star},\end{aligned} \tag{60}$$

where step (a) is derived from the Woodbury identity $(\mathbf{I}+\mathbf{A}\mathbf{B})^{-1}\mathbf{A} = \mathbf{A}(\mathbf{I}+\mathbf{B}\mathbf{A})^{-1}$ by setting $\mathbf{A} \triangleq \bar{\mathbf{H}}_i^H$ and $\mathbf{B} \triangleq \mathbf{U}\mathbf{W}\mathbf{U}^H\bar{\mathbf{H}}_i$. Hence, the stationary points $\{\mathbf{P}_{i,k}^\star\}$ of the WSR maximization problem (11) must lie in the column space of $\bar{\mathbf{H}}_i^H$.

Appendix B
Proof of Proposition 1

The matrix $\mathbf{X}_{i,k}^\ddagger$ in (23) can be rewritten by

$$\begin{aligned}\mathbf{X}_{i,k}^\ddagger &= \left(\bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H \mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H + \lambda_i \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H\right)^{-1} \\ &\quad \times \alpha_k \bar{\mathbf{H}}_i \mathbf{H}_{i,k}^H \mathbf{U}_k^\star \mathbf{W}_k^\star \boldsymbol{\Xi}_{i,k}^H.\end{aligned} \tag{61}$$

Replacing $\mathbf{H}_{i,k}^H$ with $\bar{\mathbf{H}}_i^H \mathbf{T}_{i,k}$, we have

$$\begin{aligned}\mathbf{X}_{i,k}^\ddagger &= \left(\bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H \mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H + \lambda_i \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H\right)^{-1} \\ &\quad \times \alpha_k \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H \mathbf{T}_{i,k} \mathbf{U}_k^\star \mathbf{W}_k^\star \boldsymbol{\Xi}_{i,k}^H \\ &= \left(\mathbf{U}\mathbf{W}\mathbf{U}^H \bar{\mathbf{H}}_i \bar{\mathbf{H}}_i^H + \lambda_i \mathbf{I}\right)^{-1} \alpha_k \mathbf{T}_{i,k} \mathbf{U}_k^\star \mathbf{W}_k^\star \boldsymbol{\Xi}_{i,k}^H,\end{aligned} \tag{62}$$

which is the same as $\mathbf{X}_{i,k}^\star$ in (21).

Appendix C
Proof of Proposition 2

To simplify notations, we drop the superscripts of $\mathbf{P}_i^{\text{local EZF}}$ and $\mathbf{X}_i^{\text{local EZF}}$ in this appendix. Henceforth, $\mathbf{P}_i$ and $\mathbf{X}_i$ will denote $\mathbf{P}_i^{\text{local EZF}}$ and $\mathbf{X}_i^{\text{local EZF}}$, respectively. By performing SVD on each matrix in $\{\mathbf{H}_{i,k} : k \in \mathcal{K}_i\}$, $\tilde{\mathbf{H}}_i$ is expressed as

$$\begin{aligned}\tilde{\mathbf{H}}_i^H &= [\mathbf{V}_{i,k_1}\boldsymbol{\Sigma}_{i,k_1}^H \mathbf{Q}_{i,k_1}^H, \ldots, \mathbf{V}_{i,k_{|\mathcal{K}_i|}}\boldsymbol{\Sigma}_{i,k_{|\mathcal{K}_i|}}^H \mathbf{Q}_{i,k_{|\mathcal{K}_i|}}^H] \\ &= \mathbf{V}_i \operatorname{blkdiag}\left(\boldsymbol{\Sigma}_{i,k_1}^H \mathbf{Q}_{i,k_1}^H, \ldots, \boldsymbol{\Sigma}_{i,k_{|\mathcal{K}_i|}}^H \mathbf{Q}_{i,k_{|\mathcal{K}_i|}}^H\right),\end{aligned} \tag{63}$$

with $\mathbf{V}_i \triangleq [\mathbf{V}_{i,k_1}, \ldots, \mathbf{V}_{i,k_{|\mathcal{K}_i|}}]$. Then the local EZF beamformer $\mathbf{P}_i$ in (30) can be rewritten as

$$\begin{aligned}\tilde{\mathbf{P}}_i &= \frac{\tilde{\mathbf{V}}_i(\tilde{\mathbf{V}}_i^H \tilde{\mathbf{V}}_i)^{-1}}{\|(\tilde{\mathbf{V}}_i^H)^\dagger\|_{\mathrm{F}}} \sqrt{P_{\max,i}} \\ &= \mathbf{V}_i \operatorname{blkdiag}(\mathbf{J}_{i,k_1}, \ldots, \mathbf{J}_{i,k_{|\mathcal{K}_i|}}) \boldsymbol{\Gamma}_i \\ &= \tilde{\mathbf{H}}_i^H \operatorname{blkdiag}\Big(\mathbf{Q}_{i,k_1}(\boldsymbol{\Sigma}_{i,k_1}^H)^{-1}\mathbf{J}_{i,k_1}, \\ &\qquad \ldots, \mathbf{Q}_{i,k_{|\mathcal{K}_i|}}(\boldsymbol{\Sigma}_{i,k_{|\mathcal{K}_i|}}^H)^{-1}\mathbf{J}_{i,k_{|\mathcal{K}_i|}}\Big) \boldsymbol{\Gamma}_i.\end{aligned} \tag{64}$$

Finally, by comparing $\mathbf{P}_i$ in (64) with its low-dimensional substitution structure, namely $\mathbf{P}_i = \tilde{\mathbf{H}}_i^H \mathbf{X}_i$, we have the corresponding $\bar{\mathbf{X}}_i$ expression as defined in (33).

Appendix D
Proof of Proposition 4

According to the convergence theory of BCD [34], any limit point $\left(\mathbf{U}_k^\star, \mathbf{W}_k^\star, \bar{\mathbf{P}}_{i,k}^\star, \mathbf{L}_{i,k}^\star\right)$ is a stationary point of problem (44). By rewriting the 0-1 integer constraints (40d) as $\mathbf{L}_{i,k}(m)(\mathbf{L}_{i,k}(m)-1) = 0, \forall k, \forall i, m = 1, \ldots, N_k$, we derive the Karush-Kuhn-Tucker (KKT) condition of problem (43). Upon comparing the two KKT systems of problems (43) and (44), it can be observed that any limit point of the iterative sequence generated by the RWMMSE-LSA adheres to the KKT condition of problem (43), by using the fact that the diagonal elements of $\mathbf{L}_{i,k}^\star$ must be zero or one. Given the equivalence between problem (43) and the original problem (3), which is guaranteed by the WMMSE framework [16], $(\mathbf{P}_{i,k}^\star, D_{i,k}^\star)$ corresponding to $\left(\mathbf{U}_k^\star, \mathbf{W}_k^\star, \bar{\mathbf{P}}_{i,k}^\star, \mathbf{L}_{i,k}^\star\right)$ lies in the feasible domain of the original equivalent problem (3).

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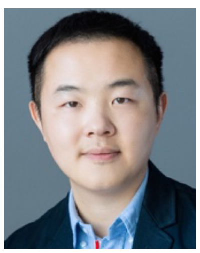

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